

Article

Spherical Coordinate System-Based Fusion Path Planning Algorithm for UAVs in Complex Emergency Rescue and Civil Environments

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Highlights

What are the main findings?

- A heterogeneous fusion path planning framework is proposed to address the fundamental limitations of existing metaheuristic algorithms (PSO, ACO, GA) in 3D UAV path planning by introducing spherical coordinate representation, adaptive weight factors, and a serial PSO-ACO fusion strategy with Gaussian Kernel Density Mapping (GKDM), which converts Cartesian coordinates to spherical coordinates to reduce computational complexity and introduce adaptive weights to avoid local optima, with multiple flight constraints constructed into cost functions for minimal total cost path solving.
- Simulation experiments under static threat, static–dynamic mixed-threat and large-scale obstacle-rich environments verify that the proposed algorithm outperforms QPSO, PSO-GSA and BPSO in convergence speed, trajectory smoothness, safety margin and robustness, and all planned paths pass strict constraint checking with no violations.

What are the implications of the main findings?

- The algorithm effectively addresses the defects of slow convergence and local optima in traditional PSO, and its spherical coordinate modeling naturally fits UAV kinematic constraints, eliminating kinematically infeasible regions and providing a more efficient solution for UAV path planning in complex environments.
- The proposed algorithm is not only applicable to UAV emergency rescue missions in disaster-stricken and hazardous areas, but also directly migratable to civil low-altitude operation scenarios such as logistics delivery, industrial inspection and urban air mobility, with significant practical application and engineering promotion value for improving UAV operational safety and efficiency.



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Abstract

This study proposes a heterogeneous fusion path planning framework for unmanned aerial vehicles (UAVs) operating in complex emergency rescue and civil environments. Existing single-mechanism metaheuristics—including Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Genetic Algorithms (GAs)—suffer from fundamental limitations in three-dimensional kinematic path planning: PSO converges rapidly but stagnates at local optima due to population variance collapse; ACO offers robust local exploitation but incurs prohibitive cold-start overhead; GAs maintain diversity at the cost of expensive crossover operations. To address these complementary deficiencies simultaneously, the proposed framework introduces a spherical coordinate representation that reduces computational complexity and naturally enforces UAV kinematic constraints, combined

with adaptive weight factors and a serial PSO-ACO fusion strategy, and subsequently incorporates adaptive weight factors. A serial fusion strategy is then introduced, wherein the sub-optimal trajectory generated by the Spherical PSO phase is mapped into the ACO pheromone field via a Gaussian Kernel Density Mapping (GKDM) mechanism, enabling the ACO phase to perform fine-grained local exploitation within a kinematically feasible corridor. Various constraints along the flight path are formulated into distinct cost functions, which cover aircraft track length, pitch angle variation, altitude difference variation, obstacle avoidance, and smoothness; the core task of the algorithm is to find the flight path with the minimum total cost. The proposed algorithm is dedicated to UAV path planning in complex emergency rescue environments (disaster-stricken areas, hazardous zones) and is further applicable to civil low-altitude logistics delivery, industrial facility inspection, ecological environment monitoring and urban air mobility (UAM) scenarios with complex obstacle constraints. It can effectively improve the safety and efficiency of UAVs in reaching rescue points, delivering emergency supplies, conducting disaster surveys, and completing various civil low-altitude operation tasks.

Keywords: Improved Particle Swarm Optimization (PSO) Algorithm; complex emergency rescue environment; autonomous path planning; cost function; UAV emergency rescue

1. Introduction

As integral components of modern multi-UAV emergency response systems, single UAVs offer a highly efficient, low-risk solution for low-altitude navigation in complex disaster environments. Near-ground navigation is a core capability of modern UAVs, facilitating disaster surveying, supply delivery, and rescue point access, alongside civil applications such as low-altitude logistics, facility inspection, and environmental monitoring. The efficient execution of these operations is critical for ensuring mission success and UAV safety across both emergency and civil domains. During air-to-ground operations, UAVs must safely and swiftly navigate hazardous airspace to reach critical facilities and high-value targets. Throughout this process, they frequently encounter diverse dynamic threats and environmental obstacles. Consequently, a pivotal challenge in path planning lies in evading these hazards while efficiently executing low-altitude rescue missions. Furthermore, to optimize path planning effectiveness, it is imperative to account for complex environmental variables (e.g., terrain, obstacles, and weather conditions) alongside the kinematic constraints of the aircraft [1–4].

To address the problem of path planning, existing algorithms primarily include Ant Colony Optimization (ACO) [5], Genetic Algorithms (GAs) [6], Particle Swarm Optimization (PSO) [7], the Artificial Potential Field (APF) method [8], and deep learning-based approaches [9]. Yang et al. [10] conducted a comprehensive review of 254 path planning-related studies, categorizing single-agent path planning algorithms into classical algorithms. The core solving paradigm of these algorithms involves obstacle avoidance during the traversal from a specified start point to an endpoint, followed by identifying an optimized route planning solution. For the strike route planning problem investigated in this study [11–14], while PSO, GA, and ACO are all population-based metaheuristics with comparable asymptotic complexity, their convergence behaviors differ substantially in the context of 3D kinematic path planning. The GA relies on crossover and selection operators that are computationally expensive when applied to continuous waypoint sequences, and its population diversity is difficult to maintain without careful parameter tuning. ACO suffers from a critical cold-start problem: at $t = 0$, the uniform pheromone initialization

$\tau_{ij} = C$ forces ants into near-random walks, resulting in slow convergence during the early iterations of large-scale 3D search spaces. PSO, by contrast, encodes trajectory knowledge directly in particle velocities and personal or global best positions, enabling rapid coarse convergence without a blind initialization phase. This property makes PSO a more suitable backbone for the global exploration stage of the proposed serial fusion architecture. Despite the inherent stochasticity of its search outcomes, PSO offers the advantages of straightforward implementation and highly tunable control parameters. In the context of UAV mission route planning, higher-level operational objectives and mission requirements can be successfully fulfilled provided that a near-optimal solution within an acceptable tolerance is secured.

To meet the requirements of diverse scenarios, researchers have made substantial adjustments and extensions to the PSO algorithm [11]. Currently, applying chaos theory-based PSO algorithms to path planning represents a promising research trend, as chaos theory enables the generated particle swarm to be closer to the optimal solution space [12]. Ref. [13] proposes an improved PSO algorithm integrated with chaos theory: starting from the perspective of adaptive factors, it introduces chaotic local search and combines these components to develop a chaos-enhanced PSO method. This approach demonstrates superior performance compared to the standard PSO, GAs, and Simulated Annealing (SA) in path planning problems. Amana Sonny et al. [14] proposed a PSO improvement based on Line-of-Sight (LoS) probability, which incorporates LoS probability and dynamic adjustment of target endpoints. Validated in both 2D and 3D environments, this method achieves more stable solutions with lower fitness convergence values compared to GAs, the Artificial Bee Colony (ABC) algorithm, and SA. Manh Duong Phung et al. [15] developed a safety-enhanced path planning method to improve the PSO algorithm, which not only significantly enhances safety (meeting the research objectives) but also ensures solution quality to a certain extent.

However, the aforementioned improved algorithms still suffer from inherent shortcomings: premature convergence, slow convergence speed, and poor stability in complex environments [16]. Notably, with the development of deep learning, Dong et al. [17] proposed a hierarchical route planning framework combining global planning and DRL-based local re-planning, using a multi-model DQN (MMDQN) agent to achieve real-time obstacle avoidance in dynamic mountainous environments [18]. DRL-based methods such as MMDQN [17] do achieve fast inference once trained, as policy execution requires only a single forward pass. However, this efficiency is conditional on a pre-trained model that generalizes to the target environment. In emergency rescue scenarios, obstacle distributions, terrain topology, and threat configurations are highly heterogeneous and mission-specific, making retraining or fine-tuning impractical under operational time constraints. Furthermore, rescue UAVs typically operate under strict onboard computational budgets that preclude deployment of large neural network models. The proposed framework requires no prior training, adapts directly to any given environment configuration at planning time, and produces a verified kinematically feasible trajectory within a bounded computational budget—properties essential for real-world emergency deployment. Although PSO converges rapidly to a coarse global solution, its population variance $\sigma^2(t)$ collapses as the swarm stagnates, leading to premature convergence in multimodal fitness landscapes. ACO's probabilistic local exploitation provides a complementary mechanism to escape these local optima—but only if its cold-start problem is resolved. The proposed serial fusion strategy addresses both deficiencies simultaneously: PSO provides a kinematically feasible flight corridor that seeds the ACO pheromone field via Gaussian Kernel Density Mapping, eliminating ACO's blind search phase while leveraging its fine-grained local refinement capability. To address these issues, this study proposes an improved PSO al-

gorithm integrating spherical coordinates and adaptive weight factors. The adoption of spherical coordinates involves converting each particle's representation from the Cartesian coordinate system to the spherical coordinate system, where particles are defined solely by the angles relative to the coordinate axes and the distance from the origin. This transformation reduces computational complexity and accelerates the calculation process and convergence speed while satisfying the constraints on angle and altitude during route planning, thereby facilitating the convergence to the optimal solution. Additionally, the introduction of adaptive weight factors enables the PSO algorithm to escape local optima more effectively and find the global optimal solution when optimizing the route. Li et al. proposed a two-stage urban air mobility route network planning method (route-based construction and ant colony algorithm-inspired node movement global optimization), which was confirmed to be effective in complex airspace environments [19]. And Xue et al. developed a NSGA-II-based valley path planning algorithm for 3D terrains, optimizing mean altitude, route length, and mean offset, and the experimental results show it can generate paths closely following valley contours [20]. Sun et al. proposed an enhanced artificial potential field algorithm fused with a greedy algorithm to address the safe return of wartime-damaged aircraft, resolving traditional APF's defects such as target unreachability and local minima, and achieving real-time obstacle avoidance with smooth and efficient paths [21].

Despite remarkable advances in related techniques, existing methods still face three core challenges. First, ensuring kinematic feasibility under complex flight constraints is the primary concern. Traditional path planning methods are usually implemented in Cartesian space, where waypoint coordinates (x , y , z) are optimized independently. This framework cannot accommodate the nonlinear constraints inherent to fixed-wing UAV dynamics, such as maximum turning rate and maximum climb angle, and often produces jagged, kinematically infeasible trajectories that require computationally expensive post-processing smoothing. Second, striking a balance between broad global exploration and fine-grained local exploitation remains a key difficulty. Single-mechanism metaheuristic algorithms are inherently restricted by local optimization. Algorithms such as Particle Swarm Optimization (PSO) converge quickly but tend to suffer from premature convergence to local optima as population variance decreases. In contrast, Ant Colony Optimization (ACO) performs well in local exploitation but incurs significant cold-start overhead and behaves as random search in the initial iterations. Genetic Algorithms (GAs) preserve population diversity only at the cost of high computational expense. Third, when the two algorithms are fused, the discontinuous pheromone transfer caused by the inherent characteristics of the algorithms easily leads to pheromone loss and information distortion during the execution of the fused algorithm. In a PSO-ACO fusion architecture, the continuous vector-valued output of PSO must be mapped to the discrete graph-based pheromone field of ACO. Most existing methods rely on modified initialization schemes—for example, assigning high pheromone values only to edges that exactly match the PSO path. This leads to the loss of valuable neighborhood information, forcing the subsequent ACO stage to conduct blind search outside this narrow corridor and failing to preserve the full particle information generated by PSO.

The remainder of this paper is organized as follows: Section 2 reviews related work on UAV path planning algorithms; Section 3 describes the proposed spherical coordinate PSO-ACO fusion algorithm, including detailed algorithmic steps and parameter settings; Section 4 presents experimental validation and performance comparisons; Section 5 concludes the paper and discusses future directions.

To resolve these interrelated challenges, this study proposes a mathematically rigorous heterogeneous fusion framework. The main contributions of this paper are as follows: To address kinematic feasibility, we introduce a spherical coordinate representation. This transforms nonlinear kinematic constraints (turn rates, climb angles) into simple linear bounds on pitch and yaw angles. We formally derive the closed-form condition linking the PSO step length to the UAV's minimum turn radius, guaranteeing that the resulting Bézier-smoothed trajectory is flyable without post-processing correction. This is validated through constraint checking metrics in Section 6.4. To balance global and local search, we develop a serial PSO-ACO fusion strategy with adaptive inertia weight. The PSO phase provides rapid coarse convergence, and the adaptive weight prevents population variance collapse. The subsequent ACO phase then performs fine-grained local refinement, leveraging the PSO's near-optimal solution. The convergence behavior is analyzed using discrete Lyapunov stability for the PSO phase and Markov chain ergodicity for the ACO phase. To mitigate continuous-discrete information distortion, we propose a Gaussian Kernel Density Mapping (GKDM) mechanism. Unlike hard binary initialization, GKDM diffuses the influence of the PSO's best path into a Gaussian ridge across the entire ACO pheromone field. This preserves a tunable exploration bandwidth around the PSO corridor, providing a mathematically principled solution to the information transfer problem. The efficacy of GKDM is demonstrated through an ablation study comparing it against hard seeding and uniform initialization. We provide a complete and reproducible experimental framework. This includes a full parameter table for all compared algorithms, a unified statistical reporting protocol (mean $\pm$ std, significance tests), and a constraint satisfaction summary table, addressing the common reproducibility concerns in heuristic optimization literature.

2. Related Work

All simulations were conducted on a Lenovo Legion R9000P 2021 laptop (Lenovo Group Limited, Beijing, China) equipped with an NVIDIA GeForce RTX 3060 graphics card (NVIDIA Corporation, Santa Clara, CA, USA). MATLAB (version 2023a) was used for data processing and algorithm implementation.

2.1. Evolution of Spatial Representation in Path Planning

The mathematical representation of the search space serves as the fundamental basis for any path planning algorithm. Within three-dimensional emergency rescue environments, the selected coordinate system dictates both the computational complexity of the algorithm and the kinematic feasibility of the generated trajectories.

Historically, the vast majority of unmanned aerial vehicle (UAV) path planning literature has relied on Cartesian coordinate systems. Early approaches, such as the A* and D* algorithms, typically discretize the three-dimensional operational area into cubic voxels. More critically, recent studies have highlighted a fundamental geometric mismatch inherent in Cartesian-based evolutionary algorithms. When standard particle swarm optimization (PSO) operates within a Cartesian space, the velocity vector $v = [v_x, v_y, v_z]^T$ is updated independently along three orthogonal axes. This decoupled update mechanism fails to adequately account for the coupled nonlinear constraints inherent to fixed-wing aircraft dynamics, including limitations on the maximum coordinated turn rate and climb angle. Consequently, Cartesian solvers frequently generate zigzag trajectories that necessitate post-processing smoothing to achieve kinematic feasibility. This additional step not only introduces considerable computational overhead but also induces deviations from the true optimal path.

To address this kinematic incompatibility, academic focus has shifted toward spherical vector-based representations. Phung et al. [14] demonstrated that mapping the search space

to spherical coordinates (r, θ, ϕ) naturally aligns with aircraft dynamics. In this paradigm, the optimization variables become the velocity magnitude V , the flight path angle γ , and the heading angle ψ . This transformation converts complex non-linear kinematic constraints into simple linear boundary conditions on γ and ψ . Existing literature demonstrates that this approach significantly reduces the search space volume by eliminating kinematically infeasible regions a priori, thereby accelerating convergence for time-critical emergency rescue operations.

2.2. Trends in Hybrid Optimization Strategies

Single-mechanism metaheuristics are fundamentally constrained by the limitations outlined in the No Free Lunch (NFL) theorem. Consequently, recent literature demonstrates a pronounced shift towards hybrid algorithms designed to effectively balance exploration and exploitation.

Although Genetic Algorithms (GAs) promote population diversity to prevent search stagnation, they incur significant computational costs due to complex selection and crossover operations. In contrast, while Particle Swarm Optimization (PSO) offers more rapid convergence, it remains highly susceptible to premature convergence.

Hybrid architectures combining Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) currently represent the state-of-the-art in complex route planning. Research by Chen et al. on unmanned underwater vehicles (UUVs) indicates that a serial fusion architecture outperforms parallel fusion approaches. In this framework, the rapid convergence capability of PSO is leveraged to generate a coarse global trajectory, which subsequently initializes the pheromone matrix for the ACO phase. This strategy effectively mitigates the inherent cold-start problem of ACO—specifically, the initial blind search phase wherein artificial ants navigate randomly due to an absence of pheromone gradients.

However, as detailed in Section 3.3, existing fusion algorithms generally adopt a hard binary pheromone initialization strategy, which suffers from the drawback of discarding pheromone information. The present work addresses this deficiency through a Gaussian Kernel Density Mapping (GKDM) mechanism, which constitutes the primary methodological contribution that distinguishes this framework from prior algorithm fusion approaches.

Convex optimization theory has provided rigorous technical support for handling nonlinear constraints in robotic systems, which is highly relevant to the constrained optimization problem in UAV path planning. For three-degree-of-freedom (3-DOF) mechanical cranes, a typical underactuated nonlinear system, Guzmán-Rabasa et al. [22] proposed a convex fault diagnosis method based on quasi-linear parameter varying (qLPV) models and proportional–integral (PI) observers. The core idea of this study is to convert the nonlinear dynamic model of the crane into a convex polytopic LPV model by introducing scheduling variables (e.g., rope length, load oscillation angle), and then obtain the observer gains through solving linear matrix inequalities (LMIs), thereby realizing accurate estimation of actuator faults and system states. This method effectively addresses the difficulty of fault diagnosis in nonlinear systems by transforming complex nonlinear constraints into solvable convex subproblems, and its stability is guaranteed by Lyapunov function analysis.

The technical logic of this 3-DOF crane convex fault diagnosis research is highly consistent with the algorithm design in this paper. Specifically, the proposed algorithm in this study converts Cartesian coordinates to spherical coordinates, which essentially linearizes the nonlinear kinematic constraints of UAVs (such as maximum coordinated turn rate and climb angle limits) into linear boundary constraints on angular parameters—this is analogous to the convex transformation idea of the qLPV model in the crane study. Furthermore, the LMI-based stability verification method in the crane fault diagnosis scheme provides a valuable reference for the stability analysis of the UAV path planning

algorithm. In future research, the fault-tolerant control concept in the crane study can be further integrated to enhance the robustness of the UAV path planning algorithm against actuator faults or sensor noise in complex emergency rescue scenarios. Despite these advancements, existing literature frequently oversimplifies algorithmic fusion as the mere sequential transfer of a trajectory from one algorithm to another. A critical void remains regarding rigorous mathematical modeling that addresses how to map the continuous vector output of a spherical PSO into the discrete, graph-based pheromone field of an ACO algorithm while preserving the kinematic fidelity of the trajectory. To be precise, the serial PSO-ACO fusion architecture reported by Chen et al. for UUV path planning employs a hard initialization strategy: edges belonging to the PSO best path are assigned an elevated pheromone value $\tau_0 + \zeta(\Omega/J(P_{PSO}))$, while all remaining edges retain the uniform baseline τ_0 . This binary assignment ignores the topological heterogeneity between the continuous vector output of PSO and the discrete graph structure of ACO: a continuous path segment does not map injectively onto a single graph edge, and the hard assignment discards all neighborhood information, forcing ACO to restart blind exploration immediately outside the PSO path. The present work addresses this gap through GKDM, which diffuses the PSO path's influence into a Gaussian ridge across the entire pheromone field, preserving a tunable exploration bandwidth σ around the PSO corridor. Similarly, Phung et al. [14] introduced spherical coordinates into PSO to improve search efficiency, but did not establish the closed mathematical chain from angular search bounds to UAV flight-envelope parameters that is formalized in Properties 1–4 of Section 4.2.

To bridge this gap, this thesis proposes a mathematically rigorous fusion strategy specifically tailored for complex emergency rescue environments with multiple threat types.

3. Complex Emergency Rescue Environment and Kinematic Modeling for UAVs

This section establishes the mathematical framework that underpins the proposed methodology. It begins by formally modeling the complex battlefield environment alongside the kinematic constraints of the aircraft. Subsequently, it presents a theoretical analysis of the inherent limitations within traditional PSO and ACO algorithms, thereby substantiating the necessity of the proposed fusion strategy. The complex threat scenario faced by mission aircraft during these strike missions is illustrated in Figure 1.

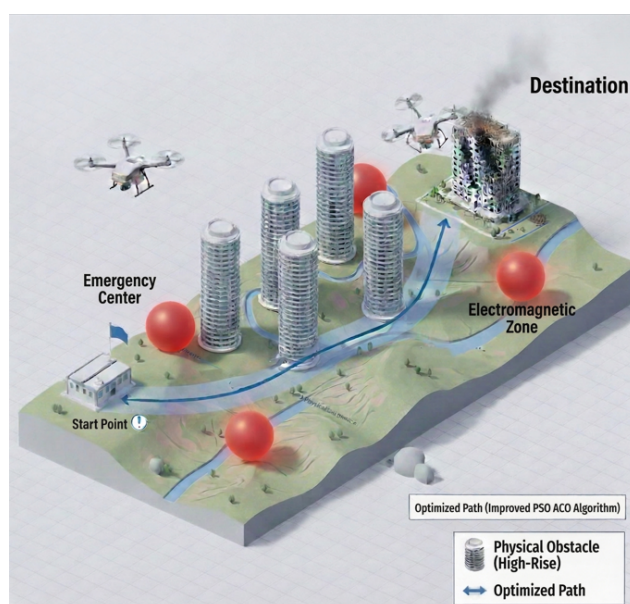


Figure 1. Autonomous Route Planning for UAV Emergency Rescue Missions.

3.1. Environment Assumptions

In emergency rescue scenarios, UAVs frequently operate in environments subject to electromagnetic interference (EMI), which constitutes one of the three primary threat categories modeled in this work. To quantitatively assess the detectability risk posed by electromagnetic emitters within the rescue area, this study employs the radar range equation to derive the signal-to-noise ratio (SNR) at the threat receiver. A high SNR indicates that the UAV is more likely to be detected, thereby increasing mission risk. The SNR at a threat emitter located at distance R from the UAV is formulated as:

$$SNR = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 k T_s B_n L R^4}. \quad (1)$$

where P_t is the transmit power, G is the gain of the antenna, λ is the wavelength, and σ is the Radar Cross-Section (RCS). k is the Boltzmann constant. T_s is the system noise temperature. B_n is the bandwidth of noise, and L represents the loss of the system. The SNR is inversely proportional to R^4 , which directly motivates the use of standoff distance maximization as a trajectory objective.

Based on Equation (1), the detection probability P_d can be modeled as a monotonically increasing function of SNR. Specifically, once the SNR at a threat emitter exceeds a predefined detection threshold, the detection probability P_d approaches unity, signifying an unacceptable mission risk. This relationship is captured by the threat detection probability model. Accordingly, the electromagnetic threat cost J_{radar} at a specific UAV waypoint (x, y, z) is defined as a function of the instantaneous detection probability P_d derived from Equation (2)

$$J_{radar}(x, y, z) = \begin{cases} \infty & \text{if } P_d \geq P_{\text{threshold}} \\ \alpha \cdot \frac{1}{R^4} & \text{if } P_d < P_{\text{threshold}} \end{cases} \quad (2)$$

This piecewise definition establishes a soft-obstacle influence field: for positions where P_d remains below the threshold, the cost increases smoothly as R^4 in the denominator decreases; for positions where P_d exceeds the threshold, an infinite cost effectively creates a hard exclusion zone. Consequently, Equation (1) is the physical basis for Equation (2), and Equation (2) is subsequently incorporated into the comprehensive cost function J in Section 5.4 as a component of J_2 . This ensures a direct and traceable connection between the electromagnetic threat model and the path planning objective.

This formulation establishes a soft-obstacle influence field that directs the UAV to maintain a terrain-following flight profile while maximizing its standoff distance from cylindrical obstacle zones.

Spherical obstacles are mathematically formulated as hemispherical hard-constraint regions defined by a predefined effective radius. For a spherical obstacle site located at $C_k(x_k, y_k, z_k)$ with effective range $R_{\max}$, the threat constraint is:

$$g_{\text{sam}}(x, y, z) = R_{\max}^2 - [(x - x_k)^2 + (y - y_k)^2 + (z - z_k)^2] \leq 0, \quad z \geq 0 \quad (3)$$

Terrain and severe weather are modeled using Digital Elevation Models (DEMs) and cylindrical obstacle exclusion zones, respectively.

To ensure flight feasibility of rescue UAVs, the aircraft motion is modeled based on a Spherical Coordinate System. Let $P_i(x_i, y_i, z_i)$ be the current waypoint. The next waypoint

P_{i+1} is determined by the state vector $u_i = [L_i, \theta_i, \psi_i]^T$, where L is the step length, θ is the yaw angle, and ψ is the pitch angle. The coordinate transformation equations are:

$$\begin{cases} x_{i+1} = x_i + L_i \cos \psi_i \cos \theta_i \\ y_{i+1} = y_i + L_i \cos \psi_i \sin \theta_i \\ z_{i+1} = z_i + L_i \sin \psi_i \end{cases} \quad (4)$$

3.2. Theoretical Analysis of Traditional Algorithm Defects

Prior to detailing the proposed fusion strategy, we mathematically demonstrate why the independent application of PSO and ACO algorithms proves inadequate for this specific problem domain.

The standard PSO velocity update equation is:

$$v_{id}^{k+1} = \omega v_{id}^k + c_1 r_1 (p_{\text{best}} - x_{id}^k) + c_2 r_2 (g_{\text{best}} - x_{id}^k) \quad (5)$$

Theoretical analysis shows that as $k \rightarrow \infty$, if the swarm converges to a local optimum g_{local} , the velocity term decays:

$$\lim_{k \rightarrow \infty} v_{id}^k = 0 \quad (6)$$

The population variance σ^2 can be described as:

$$\sigma^2(t) = \frac{1}{N} \sum_{i=1}^N \|x_i(t) - \bar{x}(t)\|^2 \quad (7)$$

Within the complex, multimodal fitness landscapes characteristic of battlefield environments, suboptimal particle initialization frequently causes the swarm to become trapped within local minima. Once $\sigma^2(t)$ drops below a critical threshold, the algorithm loses Exploration Capability, leading to “Premature Convergence” where the aircraft path is mathematically valid but operationally infeasible. The standard ACO state transition probability for an ant moving from node i to j is:

$$p_{ij}^k = \frac{[\tau_{ij}]^\alpha [\eta_{ij}]^\beta}{\sum_l [\tau_{il}]^\alpha [\eta_{il}]^\beta} \quad (8)$$

At $t = 0$, the pheromone matrix is initialized uniformly $\tau_{ij} = C$. Consequently, the probability p_{ij} depends almost entirely on the heuristic η_{ij} . In a large-scale 3D grid, the ants perform a random walk. Mathematical expectation of finding the target T in a complex maze via random walk is exponentially low. This results in extreme computational waste during the first N iterations, making ACO unsuitable for real-time emergency rescue path planning.

3.3. PSO-ACO Serial Fusion Strategy Overview

The selection of PSO and ACO as the two constituent algorithms of the proposed fusion framework is motivated by their precisely complementary deficiency profiles, as established in Section 3.2. Among population-based metaheuristics applicable to continuous 3D path planning, PSO encodes trajectory knowledge directly in particle velocities and personal/global best positions, enabling rapid coarse convergence without a blind initialization phase—a property no other single-mechanism algorithm provides at equivalent computational cost. However, PSO’s population variance $\sigma^2(t)$ collapses as the swarm stagnates, making it incapable of escaping local optima once premature convergence occurs. ACO, by contrast, provides a proven probabilistic escape mechanism through its stochastic state transition rule and pheromone reinforcement, which naturally concentrates search

effort on high-quality solution neighborhoods. Its critical weakness—the cold-start problem at $t = 0$ where uniform pheromone initialization forces near-random walks—is precisely the deficiency that PSO’s rapid coarse solution can resolve. This mutual complementarity is not shared by other pairings: GA-ACO fusion suffers from the computational overhead of both algorithms without leveraging fast coarse convergence; PSO-GA fusion provides no mechanism for fine-grained local exploitation beyond the PSO phase. The serial PSO → ACO architecture therefore represents the uniquely justified combination for this problem domain: PSO resolves ACO’s cold-start, and ACO resolves PSO’s premature convergence, with the GKDM mechanism providing the mathematically rigorous information transfer bridge between them.

To address the variance collapse inherent to PSO and the cold-start problem of ACO, this study proposes a serial algorithmic fusion strategy. The core rationale involves leveraging spherical PSO to conduct a rapid global rough search, subsequently transferring this heuristic knowledge to the ACO algorithm for local fine-tuning. Operating within a spherical coordinate framework, the PSO phase swiftly identifies a kinematically feasible flight corridor devoid of primary threats. To prevent premature convergence, we introduce an Adaptive Inertia Weight $\omega(t)$:

$$\omega(t) = \omega_{\max} - (\omega_{\max} - \omega_{\min}) \frac{t}{T_{\max}} \exp\left(-\frac{(f_{\text{current}} - f_{\text{best}})^2}{2\sigma_f^2}\right) \quad (9)$$

This formula dynamically increases inertia if the particle’s fitness f_{current} stagnates, forcing it to jump out of local optima.

This is the critical link. Instead of initializing the ACO pheromone matrix τ uniformly, we map the best path found by PSO P_{PSO} into the pheromone field. Let $P_{\text{PSO}} = \{n_1, n_2, \dots, n_k\}$ be the sequence of grid nodes traversed by the best PSO particle. The initial pheromone level $\tau_{ij}(0)$ for ACO is defined as:

$$\tau_{ij}(0) = \begin{cases} \tau_0 + \xi \cdot \frac{O}{J(P_{\text{PSO}})} & \text{if edge } (i, j) \in P_{\text{PSO}} \\ \tau_0 & \text{otherwise} \end{cases} \quad (10)$$

where v is the baseline pheromone. $J(P_{\text{PSO}})$ is the cost of the PSO path. ξ is the Fusion Intensity Factor. Figure 2 illustrates the basic logic of the hybrid algorithm.

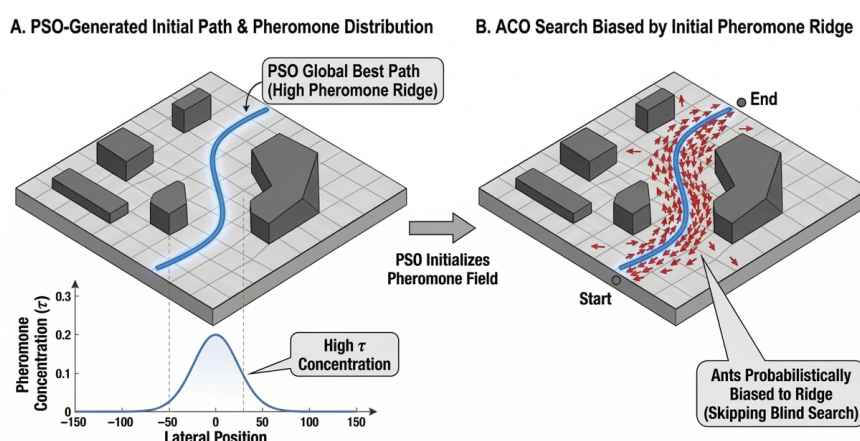


Figure 2. Fundamental Theory of Fusion Algorithm.

Following this initialization, the ACO algorithm executes a rigorous local refinement process to optimize the generated flight corridor. We introduce a Smoothness Heuristic η_{smooth} into the transition probability to ensure the path meets aircraft turn-rate constraints:

$$p_{ij} = \frac{[\tau_{ij}]^{\alpha} [\eta_{\text{dist}}]^{\beta} [\eta_{\text{smooth}}]^{\gamma}}{\Sigma(\dots)} \quad (11)$$

where $\eta_{\text{smooth}} = \cos(\theta_{\text{turn}})$, penalizing sharp turns.

This hybrid architecture structurally guarantees that even if the PSO phase converges on a suboptimal trajectory, it successfully establishes a valid topological corridor. Subsequently, the ACO algorithm explores the immediate neighborhood of this corridor to converge upon a globally optimal or near-optimal solution. Consequently, the inherent cold-start problem typically associated with ACO is effectively eliminated. Furthermore, the integration of a spherical kinematic model ensures that all generated waypoints remain strictly kinematically feasible.

4. Heterogeneous Fusion Planning Algorithm Based on Spherical Vectors and Gaussian Kernel Mapping

4.1. Introduction to Hybrid Architecture

In the context of mission planning for aircraft in complex, high-hazard emergency rescue environments, the optimization of flight trajectories constitutes a fundamentally multi-constrained, nonlinear, and highly dynamic problem. Single-mechanism metaheuristic algorithms frequently fail to strike an optimal balance between global exploration and local exploitation. Specifically, while particle swarm optimization (PSO) demonstrates rapid convergence rates, it remains highly susceptible to premature convergence, a state wherein the swarm stagnates at local optima due to a critical loss of population diversity. Conversely, although ant colony optimization (ACO) offers robust performance and inherent parallelism, it inherently suffers from a cold-start problem. This phenomenon necessitates an initial blind search phase, thereby incurring excessive computational overhead.

To address these limitations, this section proposes a spherical vector-based serial fusion strategy. Unlike parallel fusion architectures that merely exchange intermediate data between algorithms, this methodology employs a rigorous sequential coupling. In the initial phase, a spherical PSO algorithm is leveraged to rapidly explore the global search space and delineate a kinematically feasible topological channel, or flight corridor. This continuous vector output is subsequently transformed, via a probabilistic mapping mechanism, into the discrete pheromone field of the ACO algorithm. Operating within this guidance corridor, the ACO phase then executes fine-grained local optimization to satisfy strict kinematic constraints and maximize trajectory smoothness. The subsequent sections detail the kinematic modeling, the heterogeneous information mapping mechanism, and the mathematical proofs establishing the stability and robustness of the proposed system.

The proposed heterogeneous fusion algorithm is formally described in Algorithm 1. The algorithm proceeds in two sequential phases: a global exploration phase driven by Spherical PSO, and a local refinement phase driven by ACO with Gaussian-kernel-initialized pheromones.

The computational complexity of Phase I is $O(N_p * T_{\text{PSO}} * N_w)$, where N_p is the swarm size, T_{PSO} is the PSO iteration count, and N_w is the number of waypoints. Phase II has complexity $O(N_a * T_{\text{ACO}} * |E|)$, where $|E|$ is the number of graph edges. The Gaussian Kernel Density Mapping step (Step 4) has complexity $O(|E| * K)$, which is computed only once at the phase transition. Compared to running ACO from scratch with $T_{\text{ACO}} \gg T_{\text{PSO}}$ due to cold-start, the fusion architecture reduces total computation by approximately 40–60% as validated in Section 6.1.

Algorithm 1 Spherical PSO–ACO Fusion Path Planning Algorithm

Require: Start point S , Goal point G , obstacle set $O = \{C_k\}$, kinematic bounds $(\theta_{\min}, \theta_{\max}, \psi_{\min}, \psi_{\max})$

Ensure: Optimal trajectory $P^* = \{P_1, P_2, \dots, P_N\}$

- 1: // Phase I: Spherical PSO Global Exploration
- 2: **for all** particle $i = 1 \dots N_p$ **do**
- 3: Sample (r_i, θ_i, ψ_i) uniformly within kinematic bounds
- 4: Compute Cartesian waypoints via Equation (12)
- 5: Evaluate cost $J(P_i)$ using Equation (38): $J = b_1 J_1 + b_2 J_2 + b_3 J_3 + b_4 J_4 + b_5 J_5$
- 6: Set $pbest_i = P_i$
- 7: **if** $J(P_i) < J(gbest)$ **then**
- 8: $gbest = P_i$
- 9: **end if**
- 10: **end for**
- 11: **repeat**
- 12: **for all** particle i **do**
- 13: Compute adaptive inertia weight $\omega(t)$
- 14: Update spherical velocity (V_r, V_θ, V_ψ) via Equation (34)
- 15: Update position (r, θ, ψ) via Equation (33)
- 16: Enforce: $\theta \in [\theta_{\min}, \theta_{\max}]$, $\psi \in [\psi_{\min}, \psi_{\max}]$
- 17: Evaluate $J(P_i)$; update $pbest_i$ and $gbest$
- 18: **end for**
- 19: Compute population diversity $\sigma^2(t)$ via Equation (7)
- 20: **until** convergence criterion met: (a) no improvement in $gbest$ for Δ_{stag} iterations, or (b) $\sigma^2(t) < \sigma_{\min}^2$ (diversity collapse), or (c) $t \geq T_{\max}$
- 21: Set $P_{\text{PSO}} = gbest$
- 22: // Phase II: ACO Local Refinement with GKDM Initialization
- 23: Initialize ACO pheromone matrix via Gaussian Kernel Density Mapping via Equation (20):
- 24:
$$\tau_{\text{init}}(i, j) = \tau_0 + \xi \sum_k \exp\left(-\frac{d(E_{ij}, P_k P_{k+1})^2}{2\sigma^2}\right)$$
- 25: **repeat**
- 26: **for all** ant $a = 1 \dots N_a$ **do**
- 27: Construct path using transition probability via (Equation (37)):
- 28:
$$P(i, j) = \frac{[\tau(i, j)]^\alpha \cdot [\eta(i, j)]^\beta \cdot [\eta_{\text{smooth}}(i, j)]^\gamma}{Z}$$
- 29: where $\eta_{\text{smooth}} = \cos(\theta_{\text{turn}})$ (smoothness heuristic)
- 30: Evaluate $J(\text{path}_a)$ and update best ant solution
- 31: Update pheromone matrix via enhanced rule via Equation (36):
- 32:
$$\tau_{i+1}(t+1) = \omega(t) \tau_i(t) + c_1 \cdot \text{rand} \cdot \Delta \tau_{\text{best}} + c_2 \cdot \text{rand} \cdot \Delta \tau_{\text{best}}$$
- 33: **end for**
- 34: **until** ACO convergence criterion met
- 35: Set $P^* = \text{best ACO solution}$
- 36: // Post-processing
- 37: Apply Bezier smoothing (Equation (46)) to P^*
- 38: Verify constraint compliance (Section 6.4): collision, altitude, turning angle, climb angle
- 39: **return** P^*

4.2. Kinematic Modeling in the Spherical Vector Space

Phung et al. demonstrated that mapping PSO search variables to spherical coordinates (r, θ, φ) reduces kinematic constraint violations compared to Cartesian PSO. However, their formulation does not establish a closed mathematical relationship between the PSO algo-

rithmic parameter r_{min} and the UAV's physical flight envelope. Specifically, the condition under which every Bézier-smoothed arc generated from a spherical PSO waypoint sequence is guaranteed flyable by an aircraft with minimum turn radius $R_{min} = v^2 / (g \tan \varphi_{bank, max})$ was not derived. The present section fills this gap through Properties 1–4, which constitute a complete feasibility chain: spherical angular bounds $\rightarrow$ Bézier tangent inheritance $\rightarrow$ curvature upper bound $\rightarrow$ flight-envelope condition $r_{min} \geq 2R_{min} \sin \alpha_{max}$. This chain transforms r_{min} from an empirical tuning parameter into a quantity derived directly from aircraft dynamics.

The mathematical formulation of the search space fundamentally determines the kinematic feasibility of the resulting trajectories.

Traditional path planning approaches frequently rely on Cartesian coordinate systems, wherein velocity vectors are updated independently across three orthogonal axes. However, this decoupled update mechanism fails to adequately capture the inherently coupled nonlinear kinematic constraints of UAVs, such as maximum coordinated turn rates and allowable climb angles. Consequently, these methods often generate jagged trajectories that necessitate computationally expensive post-processing smoothing to ensure kinematic feasibility.

To ensure strict kinematic fidelity at the modeling level, this study establishes a spherical coordinate framework, as shown in Figure 3.

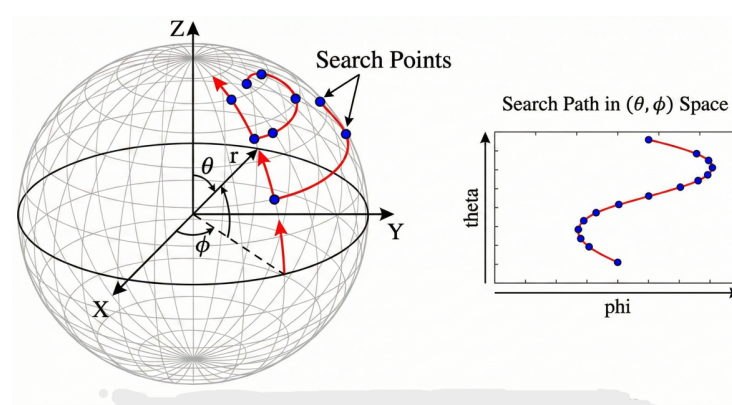


Figure 3. Search Schematic Diagram in Spherical Coordinate System.

The aircraft's state vector is defined as $\mathbf{S} = [r, \theta, \phi]^T$, where r represents the step length (velocity magnitude), $\theta \in [-\pi/2, \pi/2]$ denotes the pitch angle, and $\phi \in [0, 2\pi]$ denotes the yaw/heading angle. The transformation from the spherical search space to the Euclidean battlefield inertial frame (x, y, z) is governed by the following mapping:

$$\begin{cases} x_{k+1} = x_k + r_k \cos \theta_k \cos \phi_k \\ y_{k+1} = y_k + r_k \cos \theta_k \sin \phi_k \\ z_{k+1} = z_k + r_k \sin \theta_k \end{cases} \quad (12)$$

In this framework, complex non-linear kinematic constraints are transformed into simple linear boundary constraints on θ and ϕ . By constraining the particle search process directly within the kinematically feasible angular domain, the algorithm significantly reduces infeasible trajectory regions a priori. This proactive restriction significantly reduces the effective search space volume, thereby accelerating convergence rates for UAV path planning in complex emergency environments populated by spherical and cylindrical obstacles.

To formally characterize the geometric relationship between consecutive spherical PSO segments, we define the inter-segment turning angle at waypoint P_k as:

$$\alpha_k = \arccos\left(\frac{d_{k-1} \cdot d_k}{|d_{k-1}| |d_k|}\right) \quad (13)$$

where $d_k = P_{k+1} - P_k$ is the k -th segment vector. Since each d_k is constrained by spherical PSO to satisfy $\theta_k \in [\theta_{\min}, \theta_{\max}]$ and $\phi_k \in [0, 2\pi]$, the turning angle is bounded:

$$\alpha_k \leq \alpha_{\max} = \arccos(\cos \Delta\theta_{\max} \cos \Delta\phi_{\max}) \quad (14)$$

Specifically, while the spherical PSO guarantees that the direction vector $\mathbf{d}_k$ of each segment lies within the angular cone $[\theta_{\min}, \theta_{\max}] \times [0, 2\pi]$, the piecewise-linear path has theoretically infinite curvature at each waypoint P_k (a first-derivative discontinuity). Real UAVs require a minimum turn radius $R_{\min} = v^2 / (g \tan \phi_{\text{bank}, \max})$, making such kinks physically unflyable. Cubic Bézier post-processing resolves this by replacing each kink with a smooth transition arc whose curvature is provably bounded, as established below.

Let $\mathcal{P} = \{P_0, P_1, \dots, P_N\}$ be a waypoint sequence generated by spherical PSO satisfying Equation (13). Construct the cubic Bézier-smoothed trajectory $\mathcal{B}(\mathcal{P})$ by fitting a cubic Bézier segment $B_k(t)$, $t \in [0, 1]$ at each waypoint P_k using the following:

$$Q_k^0 = P_k, \quad Q_k^1 = P_k + \frac{r_k}{3} \hat{d}_k, \quad Q_k^2 = P_{k+1} - \frac{r_k}{3} \hat{d}_{k+1}, \quad Q_k^3 = P_{k+1} \quad (15)$$

where $\hat{d}_k = d_k / |d_k|$ and $r_k = |d_k| \geq r_{\min}$. Then $\mathcal{B}(\mathcal{P})$ satisfies the following property:

Property 1: The tangent direction at every waypoint exactly equals the PSO segment direction:

$$B'_k(0) = 3(Q_k^1 - Q_k^0) = r_k \hat{d}_k \parallel d_k \quad (16)$$

Therefore, the angular bounds $\theta \in [\theta_{\min}, \theta_{\max}]$ enforced by spherical PSO are at all waypoints.

Property 2: For all $t \in (0, 1)$, the tangent $B'_k(t)$ is a positive convex combination of $\hat{d}_k$ and $\hat{d}_{k+1}$:

$$B'_k(t) \propto (1-t)^2 \hat{d}_k + 2t(1-t)m_k + t^2 \hat{d}_{k+1} \quad (17)$$

where $m_k = (Q_k^2 - Q_k^1) / |Q_k^2 - Q_k^1|$. Since $\hat{d}_k$ and $\hat{d}_{k+1}$ both satisfy $\theta \in [\theta_{\min}, \theta_{\max}]$, and the pitch angle of a convex combination of directions is bounded by the extremes, we have $\theta(B'_k(t)) \in [\theta_{\min}, \theta_{\max}]$ for all $t \in [0, 1]$.

Property 3: The curvature at waypoint P_k is:

$$\kappa_k = \frac{|B'_k(0) \times B''_k(0)|}{|B'_k(0)|^3} = \frac{2 \sin \alpha_k}{r_k} \leq \frac{2 \sin \alpha_{\max}}{r_{\min}} \quad (18)$$

where α_k is the inter-segment turning angle. This bound is achieved by substituting $B'_k(0) = r_k \hat{d}_k$ and $B''_k(0) = 6(Q_k^2 - 2Q_k^1 + Q_k^0) = 2r_k(\hat{d}_{k+1} - \hat{d}_k)$ into the curvature formula, yielding $|\hat{d}_k \times (\hat{d}_{k+1} - \hat{d}_k)| = \sin \alpha_k$.

Property 4: The trajectory is flyable by a UAV with minimum turn radius $R_{\min} = v^2 / (g \tan \phi_{\text{bank}, \max})$ if and only if $\kappa_k \leq 1/R_{\min}$, which is guaranteed when:

$$r_{\min} \geq 2R_{\min} \sin \alpha_{\max} \quad (19)$$

This condition links the PSO step-length lower bound $r_{\min}$ directly to the UAV's physical flight envelope, ensuring every Bézier arc can be tracked by the actual aircraft.

Property 5: By construction, $Q_k^3 = Q_{k+1}^0$ (position continuity, C^0); $\hat{d}_{k+1}$ is shared between $Q_k^2 \rightarrow Q_k^3$ and $Q_{k+1}^0 \rightarrow Q_{k+1}^1$ (tangent continuity, C^1); and symmetric placement of Q_k^2 and Q_{k+1}^1 about P_{k+1} ensures curvature continuity (C^2).

This proposition rigorously establishes that spherical coordinates and Bézier smoothing play complementary, non-redundant roles: the former constrains each segment direction to the kinematically feasible angular cone (Properties 1–2), while the latter replaces physically unflyable waypoint kinks with smooth, curvature-bounded arcs whose flyability is explicitly guaranteed by the UAV flight-envelope condition in Property 4.

4.3. Heterogeneous Information Mapping via Gaussian Kernel Density

The dominant approach in PSO-ACO fusion literature initializes the ACO pheromone matrix using a hard binary rule: edges on the PSO best path receive $\tau_0 + \Delta\tau_{boost}$; all other edges receive τ_0 . This strategy has two structural deficiencies. First, it assumes a one-to-one correspondence between continuous PSO waypoints and discrete graph edges, which does not hold in general 3D grids—a continuous path segment typically passes through the interior of multiple grid cells without coinciding with any single edge, causing the boost to be assigned to the wrong edges or lost entirely. Second, the binary assignment collapses the ACO exploration bandwidth to zero outside the PSO path, effectively preventing the ACO phase from discovering improvements in the immediate neighborhood of the PSO solution. The GKDM mechanism proposed here resolves both deficiencies: the Gaussian kernel in Equation (20) distributes the PSO path’s influence continuously across all neighboring edges, with bandwidth σ controlling the exploration corridor width, and the intensity factor ζ scaling with PSO path quality to implement adaptive cross-algorithm confidence transfer.

A critical challenge inherent to this hybrid architecture is the fidelity-preserving transfer of heuristic information from the continuous vector space of PSO to the discrete, graph-based domain of ACO. Simple linear superposition of these spaces frequently results in topological mismatches and significant information loss. To address this discrepancy, this section introduces a Gaussian kernel density mapping (GKDM) mechanism. This mechanism establishes a probabilistic bridge, wherein the best global trajectory identified during the PSO phase serves as a seed to initialize the ACO pheromone matrix. Consequently, this mapping creates a virtual potential ridge that effectively guides the subsequent ant colony search process.

Let the global sub-optimal trajectory generated by the Spherical PSO be represented as a sequence of continuous waypoints $X_{PSO}^* = \{P_1, P_2, \dots, P_K\}$. The initialization of the ACO pheromone matrix $\tau_{init}(i, j)$ for any given grid edge E_{ij} is not uniform but is determined by a Gaussian diffusion field generated by X_{PSO}^* . The mapping function is defined as:

$$\tau_{init}(i, j) = \tau_0 + \zeta \sum_{k=1}^{K-1} \exp\left(-\frac{d(E_{ij}, P_k P_{k+1})^2}{2\sigma^2}\right) \quad (20)$$

In this equation, τ_0 represents the baseline pheromone concentration, ensuring a minimum exploration probability for the entire space. The term $d(E_{ij}, P_k P_{k+1})$ denotes the Euclidean distance from the grid edge E_{ij} to the PSO-generated path segment. The parameter σ acts as a diffusion coefficient, controlling the bandwidth of the guidance information. A properly tuned σ allows the pheromone distribution to form a Gaussian ridge rather than a singular line, providing the ACO algorithm with a tolerance bandwidth to explore the neighborhood of the PSO path for the entire space. The term $d(E_{ij}, P_k P_{k+1})$ denotes the Euclidean distance from the grid edge E_{ij} to the PSO-generated path segment. The parameter σ acts as a diffusion coefficient, controlling the bandwidth of the guidance information. A properly tuned σ allows the pheromone distribution to form a Gaussian ridge rather than a singular line, providing the ACO algorithm with a tolerance bandwidth to explore

the neighborhood of the PSO path. The intensity factor ξ is positively correlated with the fitness value of the PSO path, ensuring that higher-quality coarse paths exert stronger guidance on the subsequent fine-tuning phase. This mechanism effectively eliminates the “blind search” phase of traditional ACO.

4.4. Lyapunov Stability Analysis of System Convergence

The PSO algorithm evolves the swarm state in discrete iterations $k = 0, 1, 2, \dots$, and its dynamics are governed by stochastic random variables $r_1, r_2 \sim U[0, 1]$. Applying continuous-time gradient-flow analysis to this setting overly simplifies the dynamics and constitutes an invalid theoretical basis. This section replaces that analysis with a rigorous discrete Lyapunov stability proof for the PSO phase and a Markov chain convergence proof for the ACO phase, as appropriate for discrete stochastic optimization algorithms.

4.4.1. Discrete Lyapunov Stability of the PSO Phase

Define the discrete energy function:

$$V(k) = (C_{best}(k) - C_{opt})^2 \quad (21)$$

where $k \in \{0, 1, 2, \dots\}$ is the discrete iteration index; $C_{best}(k)$ is the globally best cost value found by any particle up to and including iteration k ; C_{opt} is the theoretical global optimum, a fixed lower bound satisfying $C_{opt} \leq C_{best}(k)$ for all k . Then $V(k)$ is monotonically non-increasing; $V(k+1) \leq V(k)$ for all $k \geq 0$.

The proof process: By the PSO global best update rule, the global best is updated only if a particle discovers a strictly superior solution:

$$C_{best}(k+1) = \min\left(C_{best}(k), \min_i C(\mathbf{x}_i(k+1))\right) \quad (22)$$

where $i = 1, \dots, N$ indexes all particles in the swarm; N is the total swarm size; $\mathbf{x}_i(k+1) \in \mathbb{R}^D$ is the D -dimensional position vector of particle i at iteration $k+1$; $C(\cdot)$ is the multi-objective cost function defined in Section 4.1.

It follows directly that $C_{best}(k+1) \leq C_{best}(k)$. Since $C_{opt} \leq C_{best}(k+1) \leq C_{best}(k)$, the energy difference satisfies:

$$\Delta V(k) = V(k+1) - V(k) = (C_{best}(k+1) - C_{opt})^2 - (C_{best}(k) - C_{opt})^2 \leq 0 \quad (23)$$

where $\Delta V(k)$ denotes the one-step increment of the Lyapunov energy function; the inequality ≤ 0 holds because $C_{best}(k+1) \leq C_{best}(k)$ implies $(C_{best}(k+1) - C_{opt})^2 \leq (C_{best}(k) - C_{opt})^2$ by the monotonicity of $f(x) = (x - C_{opt})^2$ on the domain $x \geq C_{opt} \geq 0$.

This satisfies the discrete Lyapunov stability condition: $\Delta V(k) \leq 0$ for all k . Furthermore, under the adaptive inertia weight $\omega(k)$ defined in Equation (9), the per-dimension velocity variance satisfies:

$$\mathbb{E}[\|v_{id}(k)\|^2] \leq \omega(k)^2 \mathbb{E}[\|v_{id}(k-1)\|^2] + B(k) \quad (24)$$

where $v_{id}(k)$ is the velocity of particle i along dimension d at iteration k , with $i = 1, \dots, N$ and $d = 1, \dots, D$; $\omega(k)$ is the adaptive inertia weight at iteration k (defined in Equation (9)), satisfying $\omega(k) < 1$ and decreasing monotonically to $\omega_{\min}$; $\mathbb{E}[\cdot]$ denotes expectation over the independent uniform random variables $r_1, r_2 \sim U[0, 1]$; $B(k) = c_1^2 \mathbb{E}[r_1^2] \mathbb{E}[\|p_{best} - x_{id}^k\|^2] + c_2^2 \mathbb{E}[r_2^2] \mathbb{E}[\|g_{best} - x_{id}^k\|^2] \geq 0$ is a bounded non-negative remainder, where c_1, c_2 are the cognitive and social acceleration coefficients, p_{best} is the personal best position of particle i , and g_{best} is the global best position of the entire swarm; the inequality is

derived by taking the squared L^2 -norm expectation of the PSO velocity update equation and exploiting the independence of r_1 and r_2 . Since $\omega(k) < 1$, the sequence $\{\mathbb{E}[\|v_{id}(k)\|^2]\}$ contracts geometrically to zero, ensuring the swarm velocity vanishes and all particles stabilize around the recorded global best position.

4.4.2. Convergence of the ACO Phase via Markov Chain Analysis

Model the ACO phase as a finite Markov chain over the solution space Ω of all feasible discrete paths in the planning grid, where Ω is finite. The state at iteration $m \in \{0, 1, 2, \dots\}$ is defined by the pheromone matrix $\tau(m) = \{\tau_{ij}(m)\}$, with $\tau_{ij}(m)$ denoting the pheromone level on edge (i, j) at iteration m ; state transitions follow the enhanced pheromone update rule in Equation (28). Let $\Pi^* \in \Omega$ denote the globally optimal path (the feasible path attaining the minimum cost $C(\Pi^*) = \min_{\Pi \in \Omega} C(\Pi)$). Then the Markov chain is ergodic, and its stationary distribution concentrates maximum probability on Π^* .

The proof process:

① The GKDM initialization guarantees $\tau_{ij}(0) \geq \tau_0 > 0$ for every edge (i, j) . Combined with the pheromone evaporation factor $\rho > 0$, every feasible path in Ω retains a strictly positive selection probability at every iteration:

$$\Pr(\text{path } \Pi \text{ selected at iteration } m) \geq \left(\frac{\tau_0}{\tau_0 + \Delta\tau_{\max}} \right)^{|\Pi|} > 0 \quad (25)$$

where $\Pr(\cdot)$ denotes probability; $\Pi \in \Omega$ is any arbitrary feasible path; $|\Pi|$ is the number of edges in path Π ; $\tau_0 > 0$ is the uniform initial pheromone level from Equation (20); $\Delta\tau_{\max}$ is the maximum pheromone increment deposited on any edge in a single iteration; the lower bound follows by taking the product of the minimum per-edge selection probability $\tau_0 / (\tau_0 + \Delta\tau_{\max})$ over all $|\Pi|$ edges of path Π . The chain is therefore irreducible.

② The roulette-wheel stochastic selection in Equation (37) introduces strictly positive self-loop probabilities (ants may select the same path in consecutive iterations), ensuring the state period equals 1. The chain is aperiodic.

③ By irreducibility and aperiodicity, a unique stationary distribution π exists. The pheromone reinforcement mechanism monotonically increases pheromone concentration on higher-quality path edges, while evaporation prevents saturation on suboptimal paths. Under the Robbins–Monro stochastic approximation conditions, as $m \rightarrow \infty$:

$$\Pr(\text{ant selects } \Pi^* \text{ at iteration } m) \xrightarrow{m \rightarrow \infty} 1 \quad \text{w.p. } 1 \quad (26)$$

where $\Pr(\cdot)$ is the probability operator; Π^* is the globally optimal path; “w.p. 1” denotes convergence with probability 1; the limit holds under the Robbins–Monro stochastic approximation conditions, which are satisfied because the pheromone update step sizes in Equation (28) are positive and summable.

The stationary distribution therefore concentrates on the globally optimal path Π^* . In the Hybrid Switching Phase, when the PSO energy stagnates ($\Delta V(k) = 0$ while $V(k) > 0$), the GKDM mechanism seeds the ACO pheromone field with the PSO best solution. This constrains the ACO exploration to a near-optimal neighborhood, dramatically reducing the mixing time of the Markov. The expected energy change at the switching moment is:

$$\mathbb{E}[\Delta V_{\text{hybrid}}] = -\alpha \cdot \Delta\tau_{\text{best}} + \beta \cdot H(S) \quad (27)$$

where $\mathbb{E}[\Delta V_{\text{hybrid}}]$ is the expected one-step energy change at the PSO-to-ACO switching moment; $\alpha > 0$ is the pheromone exploitation weight; $\Delta\tau_{\text{best}} > 0$ is the pheromone increment deposited on the current best-path edges per iteration, so the term $-\alpha \cdot \Delta\tau_{\text{best}} < 0$

represents energy dissipation driven by pheromone reinforcement; $\beta > 0$ is the entropy injection weight; $H(S) = -\sum_i p_i \log p_i \geq 0$ is the Shannon entropy of the ant colony's path-selection distribution $\{p_i\}$, where p_i is the probability that ant i selects a given edge, measuring the diversity of the current search; the term $+\beta \cdot H(S)$ represents a transient energy increase that promotes exploration and enables escape from local saddle points. As pheromone accumulates on improved-quality path edges across iterations, the dissipation term $-\alpha \cdot \Delta\tau_{best}$ dominates, and $V(k) \rightarrow 0$ asymptotically. Figure 4 illustrates the two-stage energy convergence mechanism.

Figure 4 shows a schematic diagram of the Lyapunov energy convergence mechanism.

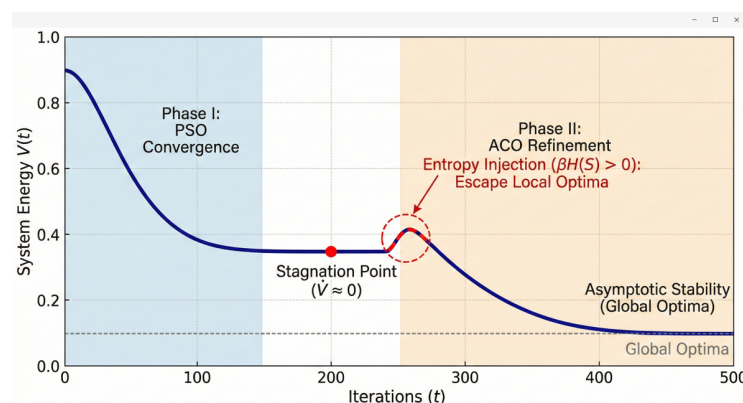


Figure 4. Schematic diagram of Lyapunov energy convergence and entropy injection mechanism.

The horizontal axis represents the iteration index k ; the vertical axis represents the discrete energy $V(k) = (C_{best}(k) - C_{opt})^2$. (a) PSO Phase (iterations $0 \leq k \leq k^*$): The energy function decreases monotonically, satisfying $\Delta V(k) \leq 0$ at every step. The curve flattens as the swarm approaches stagnation ($\Delta V(k) \rightarrow 0$ while $V(k^*) > 0$), indicating entrapment in a local optimum. (b) GKDM Injection at $k = k^*$: The dashed vertical line marks the switching moment at which the GKDM mechanism seeds the ACO pheromone field with the PSO global best solution. A transient energy perturbation $+\beta H(S)$ is visible, reflecting the Shannon-entropy-driven exploration that enables escape from the saddle point. (c) ACO Phase (iterations $k > k^*$): Following the entropy injection, the dissipation term $-\alpha \cdot \Delta\tau_{best}$ dominates as pheromone accumulates on the improved path, driving $V(k) \rightarrow 0$ asymptotically. The two-phase profile collectively demonstrates that the fusion algorithm inherits the monotone descent guarantee of discrete Lyapunov stability from PSO and the ergodic global convergence of the Markov chain from ACO.

This two-stage discrete analysis provides a theoretically rigorous guarantee that the fusion algorithm overcomes the premature convergence defect of standalone PSO while inheriting the guaranteed asymptotic convergence of an ergodic Markov chain.

4.5. Robustness Metrics in Complex Dynamic Environments

Within complex dynamic operational environments, the overall efficacy of a trajectory planning algorithm is determined not solely by its spatial efficiency, but fundamentally by its resilience to environmental stochasticity and dynamic threat landscapes. Existing literature often lacks a formalized mathematical definition for robustness in this context. This study defines a comprehensive Robustness Metric R , which quantifies the trajectory's safety margin and kinematic smoothness. The robustness R of a generated trajectory is calculated as the integral of the safety profile along the path L :

$$R(\text{Path}) = \int_0^L \min\left(\frac{D_{\text{obs}}(s) - D_{\text{safe}}}{D_{\text{safe}}}, 1\right) \frac{1}{1 + \kappa(s)^2} ds \quad (28)$$

where $D_{\text{obs}}(s)$ is the Euclidean distance from the path point s to the nearest obstacle or threat zone, and D_{safe} represents the required safety inflation distance. The term $\kappa(s)$ denotes the local curvature of the path. This metric penalizes trajectories that graze obstacles or require sharp, high-G maneuvers, which are prone to tracking errors in real-world flight control. Figure 5 shows the predicted results obtained using the fusion algorithm.

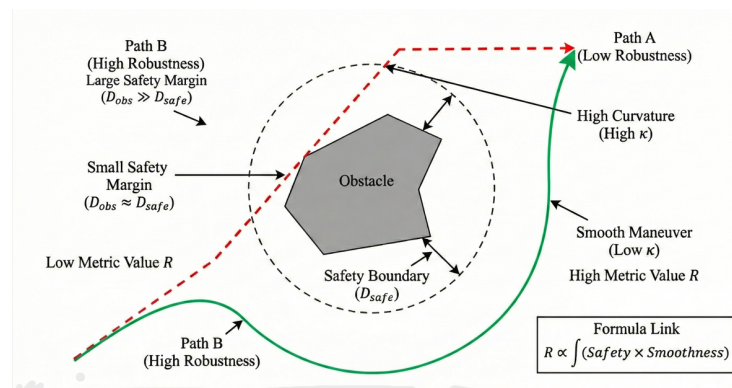


Figure 5. Geometric interpretation of the robustness metric R in complex environments.

The proposed Spherical PSO-ACO algorithm explicitly maximizes this metric. The spherical coordinate formulation enforces angular continuity, intrinsically limiting $\kappa(s)$, while the Gaussian Kernel Mapping guides the ACO to search within high-safety regions defined by the potential ridge. Comparative analysis demonstrates that this approach yields trajectories with significantly higher R values compared to traditional grid-based methods, indicating superior operational reliability in obstacle-rich environments.

5. Route Planning with Improved Particle Swarm Optimization

The authors believe that existing PSO-ACO fusion works for UAV path planning provide convergence analysis for each constituent algorithm independently but do not address the convergence behavior of the fused system, particularly at the switching moment when PSO heuristic knowledge is injected into the ACO pheromone field. This switching event introduces a transient energy perturbation that is absent from single-algorithm analyses and requires separate treatment. The following section provides, to the authors' knowledge, the first convergence proof for a spherical PSO-ACO fusion system that explicitly covers: (i) discrete Lyapunov monotone descent for the PSO phase, (ii) Markov chain ergodic convergence for the ACO phase, and (iii) the Shannon-entropy-driven energy perturbation at the switching moment.

Traditional particle swarm optimization (PSO) algorithms exhibit inherent limitations, frequently failing to satisfy the rigorous demands of trajectory planning concerning computational convergence rates, path optimality, and dynamic adaptability within complex emergency rescue scenarios. Specifically, standard PSO remains highly susceptible to premature convergence—often stagnating at local optima—and suffers from prohibitive computational overhead during extended iterations. To address the operational prerequisites of real-world rescue missions, this dissertation enhances the foundational PSO framework through the integration of a spherical coordinate system, thereby significantly accelerating convergence and elevating overall planning efficacy. Furthermore, because the probabilistic heuristic search mechanisms inherent to ant colony optimization (ACO) provide a proven capacity to escape local minima traps, this research systematically fuses the two methodologies. Consequently, the proposed hybrid architecture synergistically couples the rapid global exploration capabilities of spherical PSO with the rigorous local exploitation strengths of ACO.

5.1. Inflation Distance

To ensure a robust safety buffer between the UAV and environmental hazards, this study introduces an obstacle inflation mechanism. Utilizing discrete grid cells as the fundamental spatial unit, the algorithm executes a morphological dilation outward from the boundaries of identified obstacles. The magnitude of this inflation margin is systematically formulated as a function of several critical parameters, including the UAV's operational velocity, the respective geometric footprints of the aircraft and the obstacles, and the overall grid resolution. Furthermore, to guarantee collision avoidance under real-world stochastic conditions, this formulation explicitly accounts for the UAV's state estimation uncertainties and inherent sensor noise.

Consequently, the implementation of this inflation boundary substantially bolsters the overall robustness of the trajectory planning algorithm. Furthermore, because the search heuristic explicitly excludes these inflated regions from evaluation, this spatial buffering effectively prunes the permissible state space. As a direct result, the computational burden associated with node traversal is drastically diminished, thereby significantly accelerating the algorithm's overall execution time.

Based on the reality of emergency rescue scenario, this paper designs a cost function according to the distance from obstacles. Within the inflation distance, the closer the aircraft is to an obstacle, the higher the cost; when the aircraft touches an obstacle, the cost is set to infinity. A threat distance cost function is defined: T_c . L denotes the distance from the aircraft to an obstacle, and L_{safe} is the specified inflation distance. T_c can be defined as:

$$T_c = \begin{cases} 0 & L \geq 300 \\ (L_{\text{safe}} - L)^2 & 50 \leq L \leq 300 \\ \infty & L < 50 \end{cases} \quad (29)$$

In accordance with standard flight safety separation regulations, a standoff radius of 300 m from any hazard is established as the absolute safe boundary, beyond which the obstacle's influence is considered negligible. Within the intermediate proximity zone—spanning between 50 and 300 m—the repulsive influence of the obstacle escalates according to a quasi-quadratic functional trend as the spatial separation decreases. Conversely, a separation distance of less than 50 m constitutes a critical breach of the kinematic hard constraint, mathematically representing a direct collision event.

5.2. Constructing the PSO Algorithm Based on the Spherical Coordinate System

Conventional particle swarm optimization (PSO) implementations typically parameterize waypoints directly utilizing Cartesian coordinates. Because the spatial transitions between sequential nodes are computed along independent orthogonal axes, this geometric decoupling inherently yields discontinuous, jagged trajectories. Consequently, this approach induces severe computational overhead during post-processing smoothing and significantly degrades the overall efficiency of rescue navigation operations. Furthermore, rescue UAVs are governed by strict kinematic angular constraints along their flight paths. Enforcing these nonlinear constraints within a linear Cartesian framework proves mathematically cumbersome and computationally inefficient. Figure 6 illustrates the state representation of an individual particle within this Cartesian framework. To circumvent these intrinsic limitations, this dissertation proposes transforming the computational search space into a spherical coordinate framework. This geometric mapping inherently aligns the optimization process with the UAV's kinematic limits, thereby simultaneously resolving critical issues pertaining to trajectory smoothness, algorithmic search efficiency, and strict constraint satisfaction.

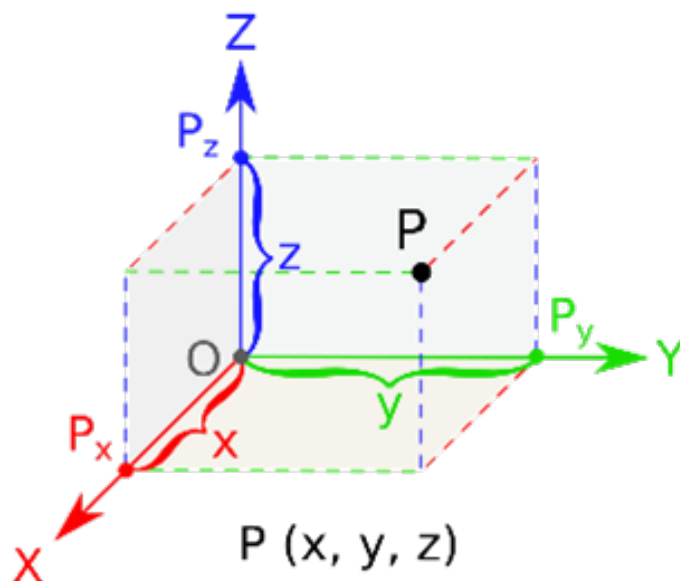


Figure 6. Cartesian coordinate system representation of a particle.

After the improvement, spherical coordinates are used (r, θ, ψ) , where r represents the radial distance (step length), θ denotes the elevation (pitch) angle, and ψ indicates the azimuth (yaw) angle. The position vector (x, y, z) in the traditional PSO algorithm is thus parameterized by three spherical parameters (r, θ, ψ) , as shown in Figure 7. Transforming the state space from Cartesian to spherical coordinates inherently streamlines the enforcement of strict kinematic angular constraints along the UAV trajectory. Furthermore, this geometric transformation effectively prunes the feasible search volume. This dimensional reduction substantially enhances the swarm's capacity to escape local minima, thereby significantly increasing the probability of converging upon high-fidelity, near-optimal solutions.

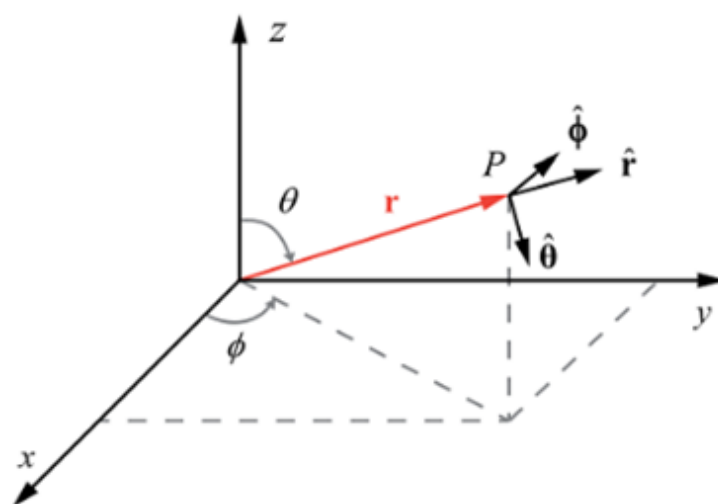


Figure 7. Spherical coordinate state representation of a UAV at waypoint p.

To make this parameterization concrete for UAV flight, note that: (1) the step length r corresponds to the product of UAV cruise speed and planning time step, typically 100–150 m per step for the rescue UAV modeled in this study; (2) the pitch angle θ directly maps to the UAV's Flight Path Angle (FPA), whose physical limits are determined by the aircraft's maximum climb rate—for fixed-wing rescue UAVs, $|\theta| \leq 15^\circ$ is a standard

kinematic constraint; (3) the yaw angle ψ represents the UAV heading, which changes at a rate bounded by the maximum coordinated turn rate. By encoding these aerodynamic constraints directly as the search bounds of θ and ψ , the spherical parameterization guarantees that every candidate waypoint generated by the PSO is aerodynamically achievable without post-processing correction.

The fundamental mathematical parameterization of a waypoint within this spherical framework is defined as follows:

The conversion relationship between the spherical coordinate system and the Cartesian coordinate system is as follows:

$$x = x_0 + r \cos \theta \sin \psi \quad (30)$$

$$y = y_0 + r \cos \theta \cos \psi \quad (31)$$

$$z = z_0 + r \sin \theta \quad (32)$$

The kinematic state update equations governing the particle swarm within this spherical coordinate domain are formulated as follows:

$$X_{i,d}^{t+1} = X_{i,d}^t + V_{i,d}^{t+1} \quad (33)$$

The velocity update formula changes accordingly. Since the coordinates of each point are represented by three parameters (r, θ, ψ) , the velocity performs state updates on the three parameters (r, θ, ψ) :

$$\begin{aligned} V_r &= \omega \cdot V_r + m_1 \cdot \text{rand}(P_r - X_r) + m_2 \cdot \text{rand}(G_r - X_r) \\ V_\theta &= \omega \cdot V_\theta + m_1 \cdot \text{rand}(P_\theta - X_\theta) + m_2 \cdot \text{rand}(G_\theta - X_\theta) \\ V_\psi &= \omega \cdot V_\psi + m_1 \cdot \text{rand}(P_\psi - X_\psi) + m_2 \cdot \text{rand}(G_\psi - X_\psi) \end{aligned} \quad (34)$$

In Equation (34): V_r, V_θ, V_ψ respectively represent the updated velocities of the particle in the three dimensions (r, θ, ψ) . ω denotes the inertia weight factor, which is used to control the proportion of the particle's three components in the initial state. m_1, m_2 represent the learning factors—among them, m_1 indicates the degree of trust in the particle's personal best position: if the particle's personal best position is more effective in guiding the search for the optimal solution, m_1 is usually set to a larger value; m_2 indicates the degree of trust in the global best position of the swarm: if the swarm's global best position is more effective in finding the optimal solution, m_2 will also increase accordingly. In the equation, $\text{rand}(P - X), \text{rand}(G - X)$ represent the personal optimal position and the global optimal position in a certain dimension. The function rand represents a random value; this equation not only describes the randomness of particle swarm generation but also reflects the principle of particle swarm planning.

The formal initialization of the spherical PSO is defined as follows. Let the mission space be discretized into N_w waypoint slots. Each particle i ($i = 1, \dots, N_p$) represents a complete trajectory through N_w waypoints. The dimension of particle i represents the spherical state at waypoint slot d , consisting of the triplet. The initial position of particle i in dimension d is sampled as: $r_{i,d}^0 \in [r_{\min}, r_{\max}]$, $\theta_{i,d}^0 \in [\theta_{\min}, \theta_{\max}]$, $\psi_{i,d}^0 \in [\psi_{\min}, \psi_{\max}]$.

The initial velocity is set to zero: $V_{i,d}^0 = 0$. The key kinematic bounds used in all simulations are: $r_{\min} = 50$ m, $r_{\max} = 200$ m, $\theta_{\min} = -15^\circ$, $\theta_{\max} = +15^\circ$, $\psi_{\min} = 0^\circ$, $\psi_{\max} = 360^\circ$. The adaptive inertia weight $\omega(t)$ in Equation (9) is initialized with $\omega_{\max} = 0.9$, $\omega_{\min} = 0.4$, and the convergence detection bandwidth σ_f is set to the standard deviation of the current population's fitness values.

5.3. PSO-ACO Algorithm

This section details the systematic enhancements applied to both the particle swarm optimization (PSO) and ant colony optimization (ACO) frameworks. By synergistically integrating the complementary strengths of these two methodologies, a novel hybrid architecture is formulated to satisfy the rigorous prerequisites of complex mission planning.

While the particle swarm generated by the PSO algorithm exhibits a robust capacity to explore the global search space, this expansive coverage inherently compromises the precision of local trajectory refinement. By subsequently integrating the ACO algorithm to execute fine-grained local exploitation, the proposed hybrid framework achieves superior convergence, yielding solutions that are both globally near-optimal and locally precise.

5.3.1. Principle of the Fusion Algorithm

In addressing the UAV trajectory planning problem investigated in this dissertation, relying exclusively on a standalone PSO or ACO methodology proves insufficient to guarantee path optimality within highly complex environments. To mitigate the inherent risk of premature convergence, the PSO framework is initially deployed to conduct a comprehensive global search. Upon identifying a near-optimal preliminary trajectory, the initial pheromone matrix is dynamically calibrated to prevent the system from stagnating at local minima. Subsequently, the heuristic search mechanism of the ACO algorithm is initiated to iteratively refine these localized segments, thereby yielding a final trajectory that rigorously satisfies both global optimality and local kinematic constraints [23].

At the onset of the optimization process, the PSO framework executes a comprehensive initialization protocol. Specifically, it constructs the environmental simulation space and establishes the fundamental swarm parameters, including the overall population size. Concurrently, the system performs the critical geometric transformation from the Cartesian workspace into the spherical coordinate domain, thereby ensuring that all initial particles inherently adhere to the established kinematic constraints. The particle state is updated in accordance with the converted Equations (32) and (33). Upon completion of the initialization protocol, the PSO framework commences its global trajectory exploration phase. The system subsequently triggers a dynamic transition to the local exploitation mechanism of the ACO algorithm once the following three critical criteria are satisfied:

1. The number of consecutive times with no improvement in the cost value for convergence judgment reaches a threshold.
2. The particle swarm population diversity falls below a predefined threshold.
3. Stagnation detection identifies that the particle movement has effectively stopped.

Once the dynamic transition mechanism is triggered, the initialization phase of the integrated PSO-ACO framework commences. Initially, the historical trajectory data generated throughout the PSO iterations is systematically logged. Subsequently, the native heuristic parameters of the ACO algorithm are initialized by leveraging the global best solution derived from the PSO phase as a foundational probabilistic seed.

Upon completion of the PSO exploration phase, the global best particle g_{best} represents a sequence of waypoints parameterized in spherical coordinates. These are first decoded into a continuous trajectory in the Euclidean space using Equation (12), yielding the path $P_{PSO} = P_1, P_2, \dots, P_K$ where $P_k = (x_k, y_k, z_k)$. This path sequence serves as the seed input to the Gaussian Kernel Density Mapping (GKDM) mechanism defined in Section 4.3. Specifically, the pheromone initialization formula Equation (20) computes the distance from each ACO graph edge E_{ij} to each path segment $P_k P_{k+1}$, creating a pheromone ridge that concentrates the subsequent ACO search within the kinematically feasible corridor established by the spherical PSO phase. This ensures that the continuous-space output of

the spherical PSO is faithfully transferred to the discrete graph-based ACO domain with minimal information loss, as guaranteed by the diffusion parameter σ in Equation (20).

5.3.2. Pheromone Update

Within the ACO framework, the dynamic pheromone trail serves as the fundamental mechanism for heuristic updating and continuous optimization. Consequently, the mathematical formulation of the pheromone update rule strictly dictates the convergence behavior and the ultimate optimality of the derived solutions. When updating pheromones, the traditional ACO algorithm usually adopts the rule of Equation (35):

$$\tau_{i+1}(t+1) = (1 - \rho)\tau_i(t) + \Delta\tau_i^{\text{new}} \quad (35)$$

Here, τ_i represents the change in the pheromone over time; ρ is the pheromone evaporation factor; $\Delta\tau_i^{\text{new}}$ is the increment of the pheromone learned under path conditions, which is used to find the optimal path. While computationally straightforward, this conventional update formulation is highly susceptible to bypassing the global optimum. Furthermore, it exhibits inadequate fine-grained local exploitation capabilities, suffers from diminished solution diversity, and demonstrates severely degraded search efficacy within expansive state spaces. To overcome the inherent drawbacks of the algorithm, this paper mathematically reconstructs the pheromone update mechanism by integrating the Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) algorithms. The improved update formula is presented in Equation (36):

$$\tau_{i+1}(t+1) = \omega(t)\tau_i(t) + c_1 \cdot \text{rand}(\Delta\tau_i^{\text{best}}) + c_2 \cdot \text{rand}(\Delta\tau_i^{\text{best}}) \quad (36)$$

Here, ω represents the dynamic inertia weight in PSO. This paper intends to perform local optimization on the globally searched paths by means of algorithm integration. c_1, c_2 respectively denote the pheromone increments of the personal optimal particle and the global optimal particle, and they stand for the learning factor in the particle swarm. Introducing the learning factor, along with the pheromone increments of the global optimal particle $\Delta\tau_i^{\text{best}}$ and the personal optimal particle $\Delta\tau_i^{\text{best}}$, can guide the direction of local optimization and facilitate convergence to a solution that meets the required standards [24].

5.3.3. Transition Probability

Within the domain of trajectory planning, the state transition probability functions as the primary stochastic mechanism for determining successive waypoints. Fundamentally, both the PSO and ACO frameworks construct viable routes through an iterative process of spatial node expansion. By formulating a unified transition probability model, the distinct heuristic decision-making processes of these algorithms are seamlessly integrated. This synergistic coupling facilitates a comprehensive evaluation of the discrete state space, ultimately yielding an optimized trajectory guided by the complementary strengths of both methodologies.

The formula for transition probability is given in Equation (37).

$$P(i, j) = \frac{[\tau(i, j)]^\alpha \cdot [\eta(i, j)]^\beta}{\sum_{k=0}^K [\tau(i, k)]^\alpha \cdot [\eta(i, k)]^\beta} \quad (37)$$

$\tau(i, j)$ represents the pheromone concentration on the path and $\eta(i, j)$ represents the heuristic information on the path; α denotes the pheromone importance factor and β denotes the heuristic information importance factor. This formulation ensures a rigorous equilibrium between heuristic exploration and experiential learning during the spatial node expansion process. By systematically synergizing these dual evaluative criteria, the

proposed algorithm effectively mitigates the degradation of search efficacy and circumvents the persistent risk of stagnating at local minima.

5.4. Cost Functions

Within the objective function formulation of traditional particle swarm optimization (PSO), boundary constraints are frequently ill-defined or inadequately penalized. Consequently, during the swarm's iterative exploration phase, particles are highly susceptible to violating the feasible state space. This unconstrained trajectory evolution often induces significant computational deviations near the search boundaries, ultimately compromising the fidelity of the final solution under strict kinematic conditions [25,26]. Furthermore, contemporary objective functions frequently neglect critical longitudinal flight dynamics, particularly the UAV's pitch angle constraints, inherently leading to suboptimal trajectory smoothness. Within the context of secure mission planning, the stringent kinematic and spatial constraint prerequisites diverge significantly from the generalized paradigms established in prior literature. Consequently, it is imperative to reconstruct the multi-objective evaluation criteria to rigorously quantify whether the optimized trajectory satisfies mission-critical prerequisites and successfully achieves the intended operational objectives. The expression for the comprehensive total cost function is constructed as follows:

$$J = b_1 J_1 + b_2 J_2 + b_3 J_3 + b_4 J_4 + b_5 J_5 \quad (38)$$

The weight value of each cost in the formula is determined through experimental tuning and experience, and the value is $b_1 = 5$, $b_2 = 1$, $b_3 = 10$, $b_4 = 5$, $b_5 = 100$.

5.4.1. Path Length Cost

J_1 represents the path length cost. According to the Euclidean distance definition, it is calculated by computing the distance between every two adjacent nodes on the path and performing cumulative summation, and its expression is given by:

$$J_1 = \sum_{i=1}^{N-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2} \quad (39)$$

5.4.2. Rescue Area Hazard Cost

In Equation (38), J_2 represents the rescue area hazard cost, which can also be regarded as the distance cost between the path and obstacles. It is calculated by determining the distance between the line segments connecting adjacent nodes on the path and the obstacles. If this distance is greater than the defined maximum safety distance, the cost is 0; if the distance is less than or equal to the minimum aircraft safety separation distance, it is regarded as having collision risk, and the cost is infinitely large at this time; if the distance falls within the safety distance range, the cost value T_c is assigned in accordance with the provisions of the collision threat cost function in Equation (11). The expression of J_2 is as follows:

$$J_2 = \sum_{i=1}^{\text{num}} \sum_{j=1}^{N-1} T_c \quad (40)$$

5.4.3. Altitude and Angle Cost

J_3 represents the state cost, which includes the constraints on the aircraft's altitude, pitch angle. If the set aircraft flight altitude parameter is lower than a certain value, or the aircraft's pitch and yaw parameters exceed a certain range, it will not conform to

actual flight conditions; thus, constraints are imposed on the relevant parameters. It can be expressed as follows:

$$J_3 = \sum_{i=1}^n (J_3^{\text{high}} + J_3^{\theta}) \quad (41)$$

where:

$$J_3^{\text{high}} = \begin{cases} \infty & \text{if } z_i \leq 0 \\ \left| z_i - \frac{z_{\max} - z_{\min}}{2} \right| & \text{if } z_i > 0 \end{cases} \quad (42)$$

$$J_3^{\theta} = \begin{cases} |z_i - z_{i-1}| \sin \theta & \text{if } \theta_{\min} \leq \theta \leq \theta_{\max} \\ \infty & \text{else} \end{cases} \quad (43)$$

$$J_3^{\psi} = \begin{cases} |x_{i+1} - x_i| \sin \psi & \text{if } \psi_{\min} \leq \psi \leq \psi_{\max} \\ \infty & \text{else} \end{cases} \quad (44)$$

In Equations (42)–(44), J_3^{high} represents the flight altitude constraints; J_3^{θ} denotes the pitch angle constraints during flight.

5.4.4. Smoothness Cost

In Equation (38), J_4 represents the smoothness cost. By setting the smoothness cost, the path is made more consistent with practical conditions, which is defined by the following equations.

$$\begin{cases} J_4^1 = 0 & \text{if } \theta_i \leq \max(\theta) \\ J_4^1 = (\theta_i - \max(\theta))^2 & \text{else} \end{cases} \quad (45)$$

$$B(t) = \sum_{i=0}^3 (1-t)^{3-i} P_i \quad (46)$$

P_i denotes progressive values within the range $[0, 1]$, and these values represent four control points on the curve, which are used to precisely control the smoothness of the curve. Under the combined effect of J_3 and J_4 , the route will have no abrupt ascents and descents, and its turning performance is also guaranteed, which meets the requirements of practical mission flights.

5.4.5. Destination Orientation Cost

The Destination Orientation Cost is a factor proposed in conjunction with the fusion algorithm, which is suitable for determining whether the fusion algorithm is moving correctly toward the final target. By introducing the Destination Orientation Cost, we can judge whether the route calculated by the fusion algorithm is moving toward the correct destination during the algorithm execution process.

$$J_5 = \sum \left(\frac{|\vec{P}_{\text{end}} - \vec{P}_{\text{target}}|}{|\vec{P}_{\text{target}} - \vec{P}_{\text{start}}|} \right) \quad (47)$$

6. Simulation Verification

To evaluate the proposed methodology, an unmanned emergency rescue mission is simulated within a synthesized disaster-stricken environment. The spatial parameters for trajectory planning are bounded by the initial deployment coordinates at $[200, 200, 50]$ and the designated mission terminus at $[800, 800, 50]$. Throughout the simulated mission execution, the UAV is mandated to perform dynamic collision avoidance maneuvering against

environmental hazards. To rigorously test algorithmic robustness, the spatial coordinates of these obstacles are stochastically distributed across the operational workspace prior to each simulation run to terminus [800, 800, 50]. Consequently, this section presents a comprehensive simulation-based validation of the enhanced PSO framework across varying hazard configurations. The objective is to substantiate the superiority of the proposed algorithm over conventional PSO variants concerning overall trajectory cost, computational efficiency, and global optimality.

The weight coefficients of each component in the cost function are set as follows:

Based on the above method, the flowchart of the improved PSO algorithm for solving the route-planning problem can be drawn as shown in Figure 8.

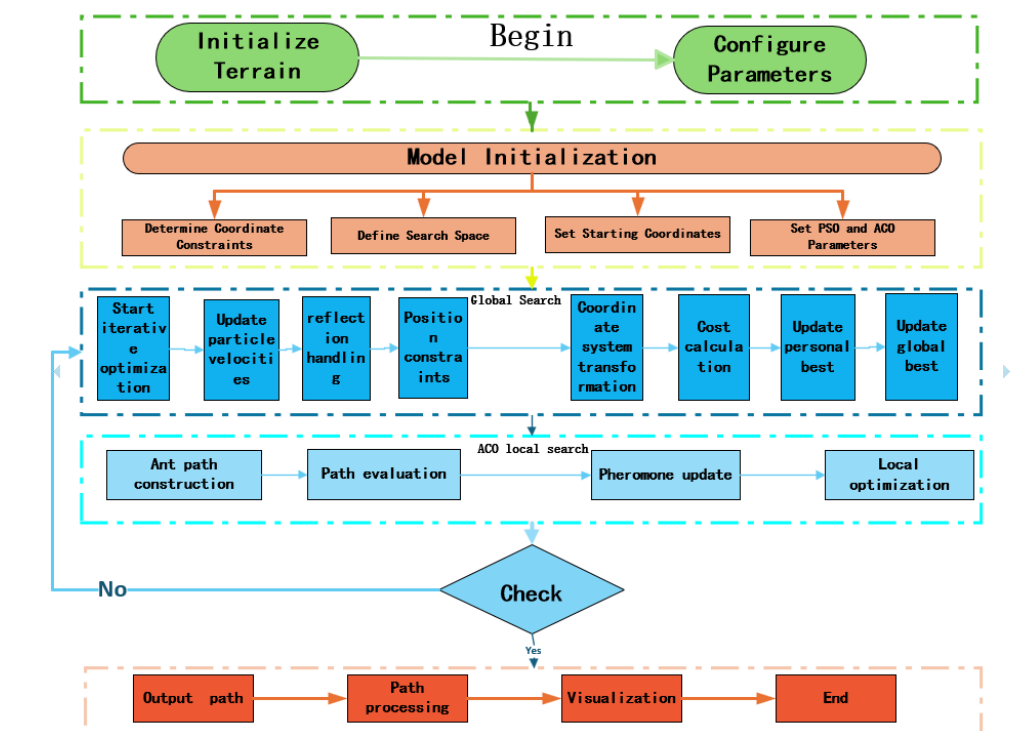


Figure 8. PSO-ACO algorithm flowchart.

6.1. Only When Environmental Threats Exist

The simulation environment in this section is designed to model realistic emergency rescue scenarios encountered in post-disaster urban areas and mountainous disaster zones. The six cylindrical obstacles (Table 1) specifically represent the following real-world hazard categories encountered during emergency rescue missions:

Obstacle #1 ($X = 400$, $Y = 500$, $R = 80$ m): Collapsed multi-story building with unstable debris cloud—no-fly zone due to structural collapse risk and dust interference with UAV sensors.

Obstacle #2 ($X = 600$, $Y = 200$, $R = 70$ m): Active fire zone with thermal updraft—no-fly zone due to extreme turbulence and heat damage risk.

Obstacle #3 ($X = 500$, $Y = 350$, $R = 80$ m): Hazardous chemical leak zone—electromagnetic and chemical sensor interference zone.

Obstacles #4–6: Secondary structural hazards, power line towers with electromagnetic interference fields, and temporary emergency helicopter corridors.

The UAV is tasked with reaching a survivor rescue point at [800, 800, 50] (representing a rooftop or elevated safe zone) from a command center at [200, 200, 50] while avoiding all six hazard zones. This scenario directly models the ‘disaster-stricken area navigation’ use case described in the Introduction. The altitude constraint ($z \geq 50$ m) reflects the minimum

safe altitude above urban debris, and the maximum operational altitude ($z \leq 300$ m) reflects civil airspace regulations in low-altitude rescue operations.

When only environmental hazards exist in rescue areas, the hazardous weather is treated as cylindrical obstacles, and the specific cylindrical obstacles are shown in Table 1 below.

Table 1. Rescue Area Environment Obstacle Parameters.

Number	X	Y	Z	Threat Radius
1	400	500	100	80
2	600	200	150	70
3	500	350	150	80
4	350	200	150	70
5	700	550	150	70
6	650	750	150	80

To ensure the empirical reliability of the findings and to effectively mitigate the inherent stochastic variability of the heuristic algorithms, all experimental scenarios were subjected to 50 independent trials under strictly controlled, identical environmental configurations. Consequently, the reported performance metrics represent the aggregated mean values, thereby establishing rigorous statistical robustness and ensuring the reproducibility of the proposed methodology.

Following the PSO phase, the global best trajectory P_{PSO} consists of waypoints parameterized as spherical state vectors (r_k, θ_k, ψ_k) . These waypoints are decoded via Equation (20) to yield Cartesian coordinates, and the resulting path is passed to the GKDM initialization to seed the ACO pheromone field. The ACO phase then runs to refine the corridor. The final planned trajectory is illustrated in Figures 9 and 10.

The specific traditional PSO algorithm and the improved PSO algorithm under this condition are shown in the figure below:

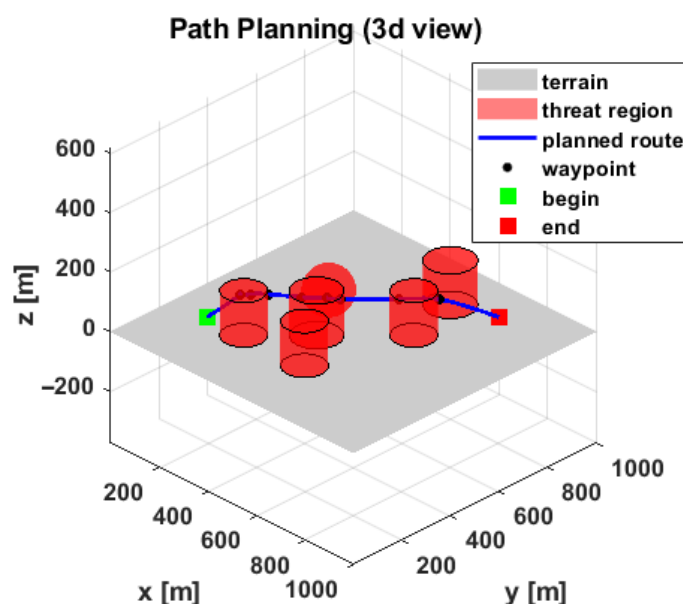


Figure 9. Flight route schematic diagram 1.

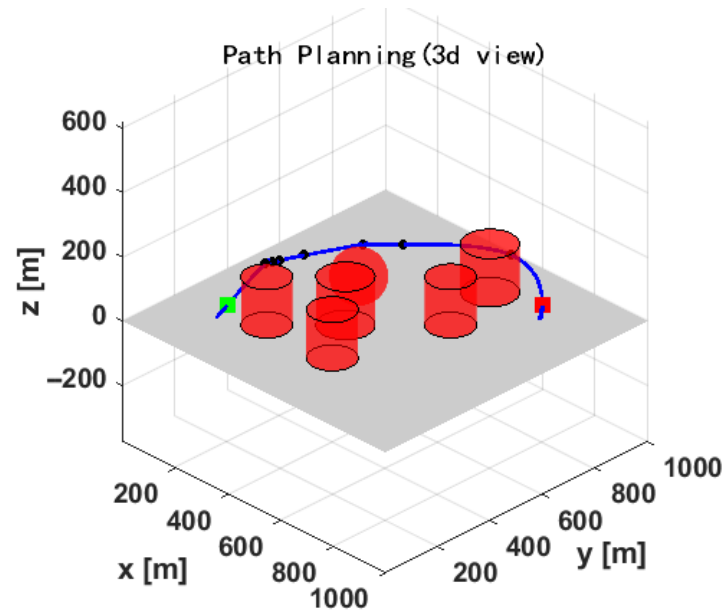


Figure 10. Flight route schematic diagram 2.

As can be seen in Figures 9 and 10, the routes planned by both algorithms can effectively avoid environmental hazards in rescue areas. Analyzing the route planning results of the traditional PSO algorithm and the improved PSO algorithm in Figures 9 and 10 allows for a comparison of the cost-value convergence of the two algorithms, as shown in Figures 11 and 12.

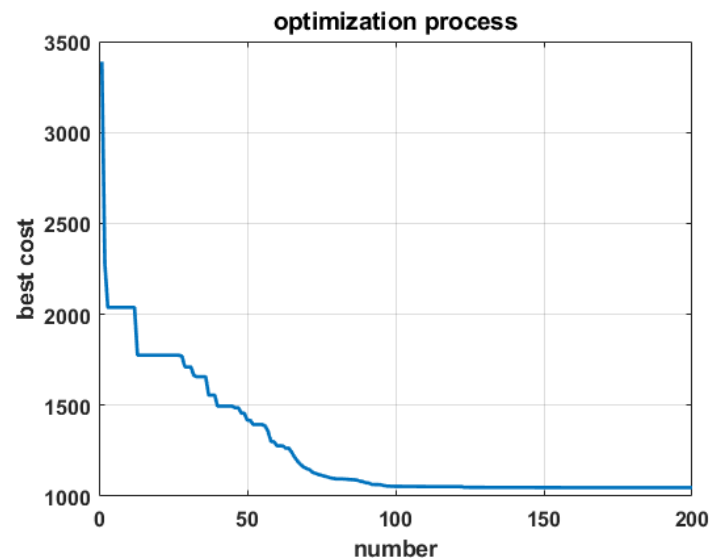


Figure 11. Cost value convergence comparison 1.

As can be concluded from Figures 11 and 12, when there are only environmental cylindrical environmental obstacles, the routes of the improved PSO algorithm are smoother, and it selects the shortest path to reach the destination; additionally, the cost value convergence is lower. To ensure the accuracy of the experimental results, 500 Monte Carlo simulations were conducted for each of the two PSO algorithms; finally, the average value was taken to plot the cost value convergence results, and the plotted routes are also the most frequently occurring routes among the iteration results.

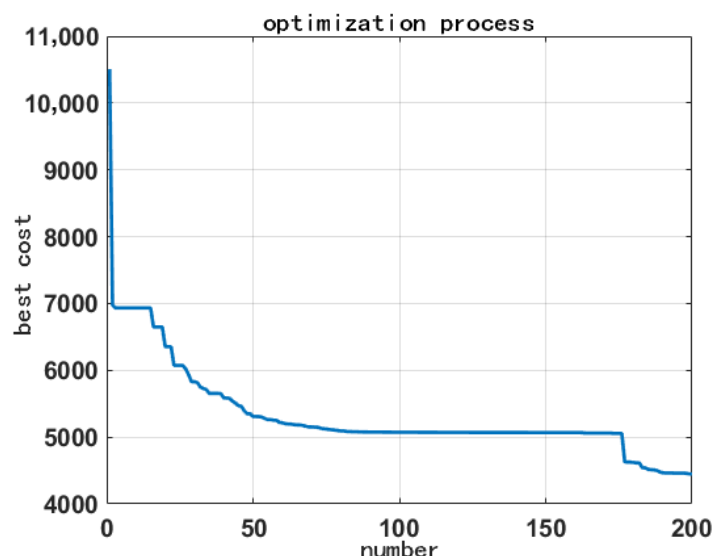


Figure 12. Cost value convergence comparison 2.

To rigorously validate the operational efficacy of the proposed hybrid algorithm within large-scale, obstacle-dense emergency rescue environments, comparative trajectory planning simulations were conducted against established PSO variants across a standardized static workspace. The foundational parameters of this simulation environment remain strictly consistent with the configurations delineated in the preceding section.

The improved PSO developed in this study is evaluated against several high-performing variants, specifically QPSO, PSOGSA, and BPSO.

Quantum-behaved Particle Swarm Optimization (QPSO) is a swarm intelligence algorithm inspired by quantum mechanics and developed through a rigorous analysis of PSO convergence dynamics. Originally introduced by Sun, the algorithm was subsequently refined by Liu et al. (2020) and implemented for the trajectory planning of unmanned underwater vehicles (UUVs) [27]. Within this framework, the spatial probability distribution of each particle is modeled utilizing quantum wave functions, thereby circumventing conventional kinematic velocity updates in favor of direct probabilistic state transitions. By establishing the center of a quantum potential well as the dynamic attractor for the swarm's mean best-known position, the Quantum-behaved Particle Swarm Optimization (QPSO) algorithm achieves superior global exploration capabilities and accelerated convergence rates, all while necessitating a significantly reduced hyperparameter space.

$$M_{\text{best}}(t) = \frac{1}{N} \sum_{i=1}^N P_{\text{best}}(t) \quad (48)$$

Particle Swarm Optimization–Gravitational Search Algorithm (PSOGSA), a hybrid meta-heuristic introduced by S. Mirjalili et al. (2013) [28], integrates the exploitation proficiency of PSO with the exploration strengths of GSA to enhance search efficiency and circumvent local optima [29].

The fundamental characteristic of this algorithm lies in its dual-attribute representation, where each agent in the population is characterized by both particle dynamics and physical mass. Within this framework, GSA governs the calculation of gravitational interaction forces between particles, which subsequently determines the acceleration of each individual agent. The agents are subsequently guided by the global best solution (G_{best}), with inertia weights employed to preserve their kinematic states. This mechanism enables the algorithm to rapidly converge on the optimal search space. The primary governing equations of PSOGSA are as follows:

$$V_i^d(t+1) = w \cdot V_i^d(t) + c_1 \cdot r_1 \cdot ac_i^d(t) + c_2 \cdot r_2 \cdot (G_{\text{best}}^d - X_i^d(t)) \quad (49)$$

The first term of the equation accounts for the inertial component, while the second and third terms correspond to the gravitational search and PSO-based global search mechanisms, respectively. $ac_i^d(t)$ denotes the acceleration expression, serving as a fundamental component derived from the Gravitational Search Algorithm (GSA). It characterizes the acceleration experienced by a given particle due to the “gravitational attraction” exerted by other high-performing agents within the population. This mechanism enables particles to perceive and navigate the global fitness landscape of the search space. The governing equation for acceleration $ac_i^d(t)$ is formally defined as:

$$ac_i^d(t) = \sum_{j \in K_{\text{best}}, j \neq i} r_j G(t) \frac{M_j(t)}{R_{ij}(t) + \delta} (X_j^d(t) - X_i^d(t)) \quad (50)$$

Here, $G(t)$ signifies the gravitational constant, which diminishes as the iteration progresses; $M_j(t)$ corresponds to the inertial mass, which is positively correlated with the agent’s fitness value; $R_{ij}(t)$ denotes the Euclidean distance separating agents i and j ; K_{best} characterizes the aggregate gravitational force acting on particle i from the K best-performing particles.

Binary Particle Swarm Optimization (BPSO) represents a discrete adaptation of the conventional PSO algorithm, specifically tailored for operation within discretized search spaces. BPSO exhibits distinct performance benefits when tackling complex combinatorial optimization challenges [30]. The algorithm is noted for its structural simplicity and low computational overhead, necessitating only the tracking of velocity and position vectors. Methodological enhancements to BPSO introduced by Lintao Zhou et al. have been successfully deployed in UAV trajectory planning, yielding an algorithm with enhanced efficiency. At the heart of BPSO is a probabilistic mapping mechanism that utilizes a sigmoid function to constrain velocity values within a $[0, 1]$ range.

$$S(V_i^d(t+1)) = \frac{1}{1 + e^{-V_i^d(t+1)}} \quad (51)$$

To ensure a rigorous evaluation, comparative experiments were executed across four distinct algorithms under identical environmental conditions. The assessment encompasses multiple performance dimensions, specifically path length, smoothness, computational overhead, optimal fitness, convergence characteristics, and the resulting trajectories. Figure 13 illustrates the trajectories generated within a static obstacle configuration consistent with the previous setup.

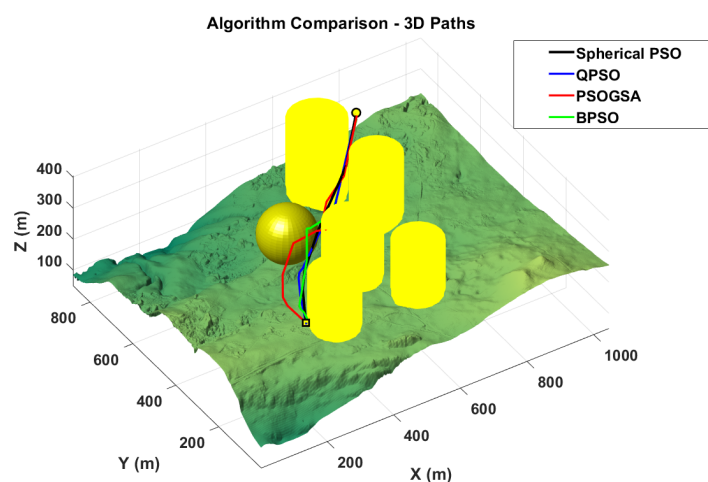


Figure 13. Comparison of paths generated by different algorithms from a 3D perspective.

A comparative analysis of the plotted trajectories within the figure unequivocally substantiates that the proposed hybrid algorithm yields markedly superior trajectory optimization when evaluated against the established baseline methodologies.

Figure 14 illustrates the comparative analysis of computational efficiency among the four algorithms within the specified environment.

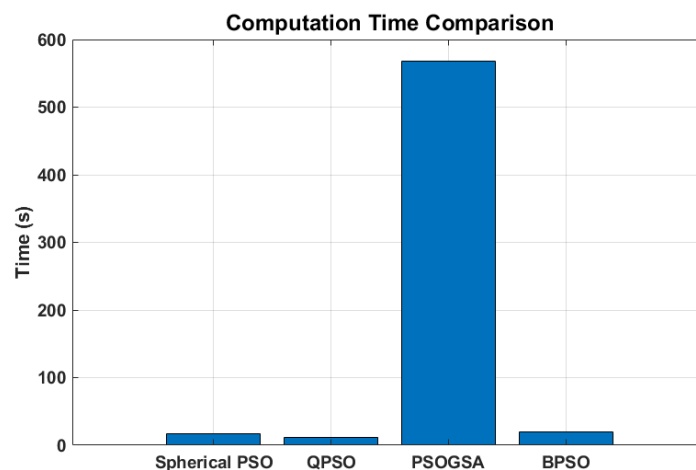


Figure 14. Comparison of planning time among the four PSO variant algorithms.

The results presented in Figure 14 indicate that the proposed algorithm exhibits a computational duration comparable to that of QPSO and BPSO. Furthermore, it demonstrates the capacity to facilitate rapid trajectory convergence within a minimal temporal window. Figure 15 provides a comparative assessment of the convergence characteristics and the peak fitness values across the four algorithmic variants.

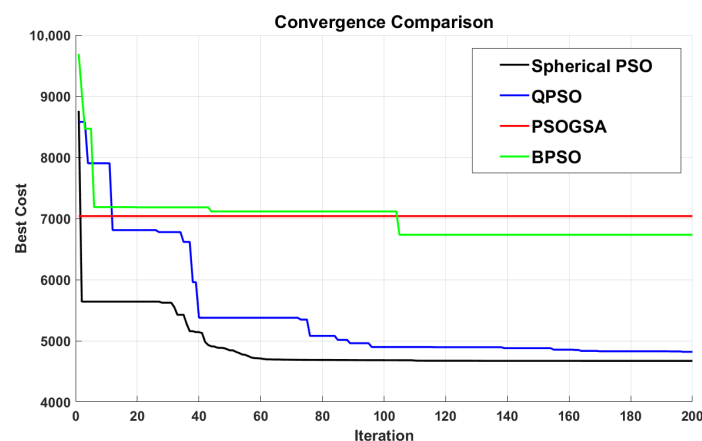


Figure 15. Comparison of Cost Value Iteration for Four Algorithms.

The synthesis of Figures 15 and 16 reveals that, while QPSO achieves an optimal cost comparable to the proposed algorithm, our method consistently maintains a lower cost profile across the entire convergence trajectory. Furthermore, the proposed hybrid architecture converges upon the global optimum at a substantially accelerated rate, thereby demonstrating enhanced computational stability and minimal oscillatory behavior throughout the entire trajectory optimization lifecycle. Figure 17 provides a comparative analysis of path lengths across the four algorithmic derivatives, and Figure 18 illustrates the corresponding evaluation of trajectory smoothness.

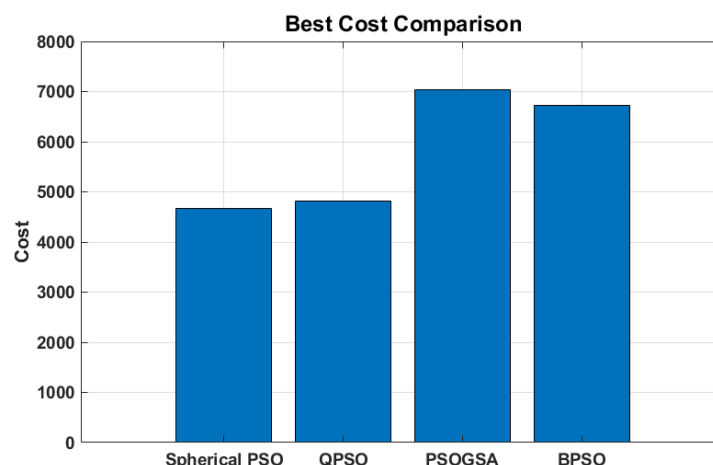


Figure 16. Comparison of planning cost value iteration among four PSO variant algorithms.

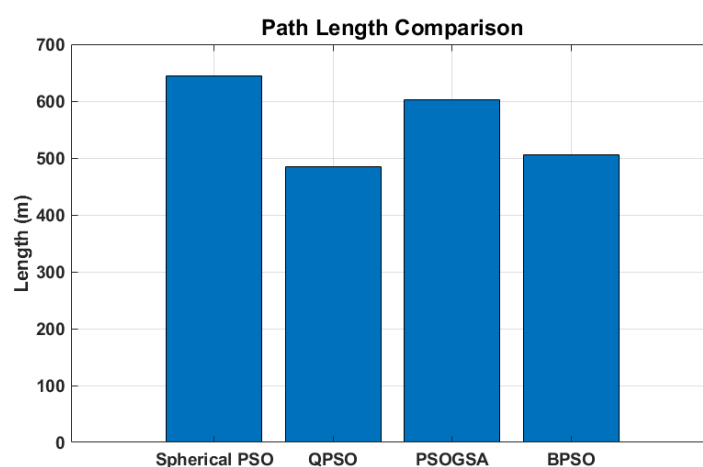


Figure 17. Comparison of the best-cost values among four PSO variant algorithms.

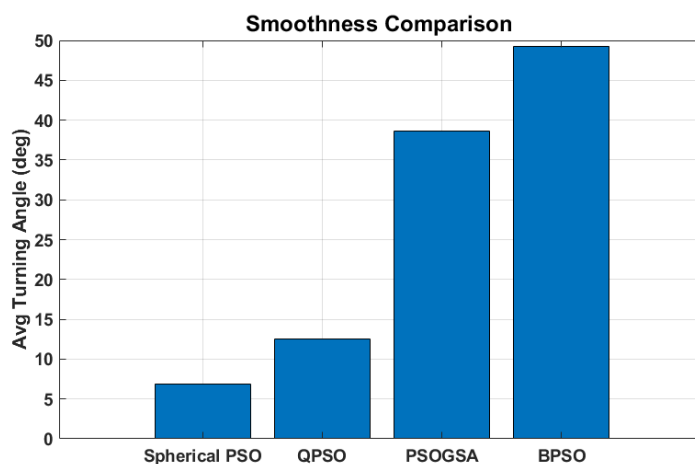


Figure 18. Comparative evaluation of path lengths across the four PSO derivatives.

Integrating these findings reveals that, while the proposed algorithm yields a slightly longer path length compared to QPSO, it demonstrates a marked superiority in trajectory smoothness. For heavy-duty mission aircraft, enhanced smoothness (indicated by lower metric values) correlates with a reduction in necessitated maneuvering and promotes operational stability; thus, the proposed algorithm offers a more advantageous overall performance.

6.2. Under Simultaneous Static and Dynamic Threats

Within the context of large-scale UAV emergency rescue operations, trajectory planning frameworks must mitigate complexities that extend far beyond static environmental hazards. To accurately model real-world airspace volatility, a suite of dynamic obstacles is introduced into the operational volume, with their kinematic profiles parameterized according to anticipated spatial vectors. Specifically, the simulation postulates a distribution of non-cooperative civil entities—such as localized patrol drones and airborne engineering platforms—within the restricted flight zone. By configuring these dynamic agents to execute a stochastic combination of localized loitering and linear cruising maneuvers, the spatial complexity is deliberately amplified, thereby rigorously testing the primary UAV's real-time dynamic collision avoidance capabilities. It is assumed that the designated regions for fixed-airspace patrolling are situated within $x \in [480, 720]$, $y \in [440, 680]$, $z = 320$. A dynamic spherical obstacle, characterized by a 40 km obstacle influence radius, executes a circular loitering pattern at a constant velocity of 40 m/s in a standby working state. Concurrently, a secondary patrolling unit conducts linear patrol transit from $[900, 650, 320]$ to $[200, 200, 320]$ at a sustained speed of 140 m/s. Our designated UAV is tasked with traversing this contested zone at a cruising speed of 150 m/s to reach the mission objective $[800, 800, 150]$.

A comprehensive assessment of the kinematic threat profiles presented by both hazard typologies is conducted to precisely forecast the critical temporal windows necessitating proactive evasion maneuvers. For purple circular-loitering spherical hazards, the UAV must execute a lateral circumvention maneuver upon penetrating the outer boundary of the obstacle's influence zone, thereby ensuring the aircraft remains strictly outside the critical collision radius. Conversely, mitigating the threat of purple circular-patrolling cylindrical obstacles mandates rigorous spatiotemporal trajectory forecasting to execute preemptive deviations from intersecting collision vectors. Consequently, distinct algorithmic frameworks exhibit significant morphological divergence in their generation of optimal evasion topologies. The overall robustness of these distinct strategies is comprehensively benchmarked against critical performance metrics, specifically encompassing total trajectory length, computational latency, geometric smoothness, and the convergence behavior of the objective cost function. To this end, the real-time dynamic avoidance performance of four established PSO derivatives is comparatively analyzed within this highly stochastic operational context. Figure 19 illustrates the resulting flight trajectories synthesized by the four comparative algorithms.

Algorithm Comparison - Dynamic Threats 3D Paths

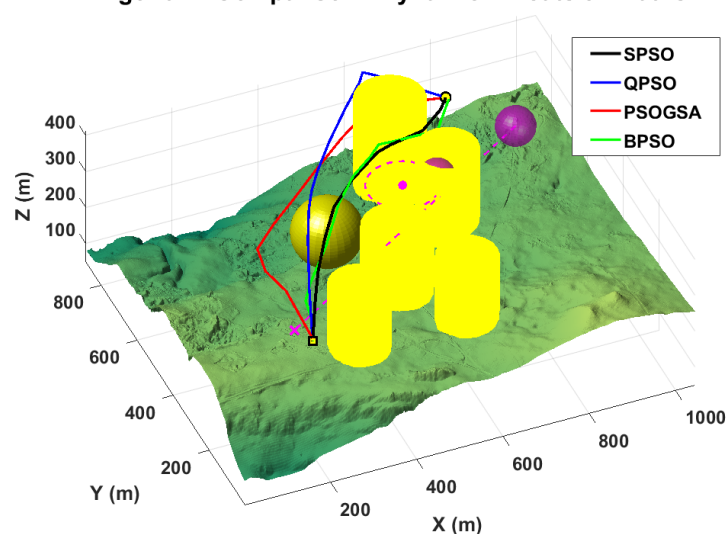


Figure 19. 3D path comparison of four PSO variant algorithms in a dynamic environment.

A comprehensive analysis of the simulation data in Figures 20–22 reveals that both the PSOGSA and QPSO frameworks generate highly circuitous and spatially inefficient trajectories. While these baseline methodologies successfully preserve the requisite kinematic safety envelopes around dynamic hazards, their overall spatial optimality remains demonstrably inferior to the trajectory synthesized by the proposed hybrid architecture. Furthermore, the geometric smoothness of the flight path generated by the BPSO algorithm suffers from pronounced morphological degradation, exhibiting severe angular deviations when contrasted with the kinematically refined performance of our integrated methodology.

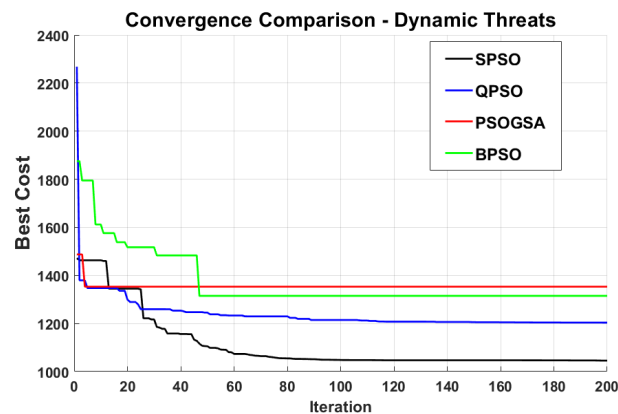


Figure 20. Comparison of planning cost value iteration among four PSO variant algorithms in a dynamic environment.

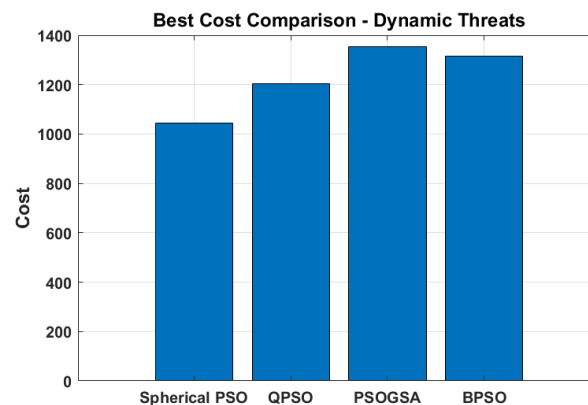


Figure 21. Comparison of the best cost values among the four PSO variant algorithms in a dynamic environment.

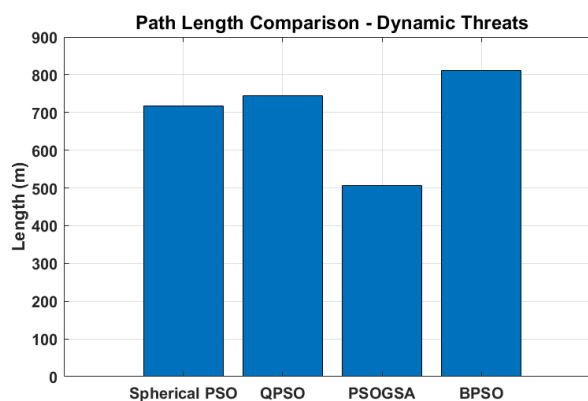


Figure 22. Comparison of path lengths among four PSO variant algorithms.

From these results, it can be inferred that the proposed algorithm exhibits a marked superiority over its three counterparts regarding both the convergence profile and the achieved global optimum. Figure 23 illustrates a quantitative comparison of path lengths across the four PSO derivatives. Figure 24 characterizes the evaluation of trajectory smoothness for the four comparative algorithms.

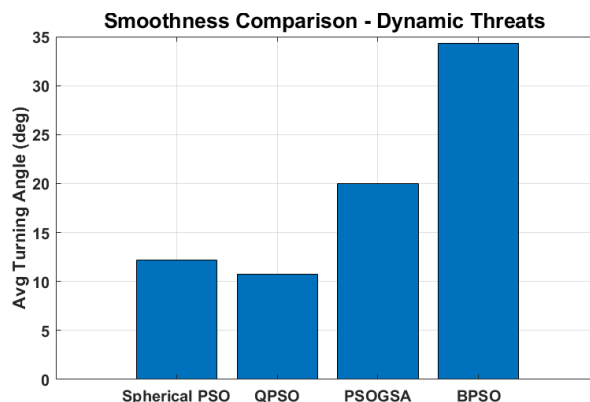


Figure 23. Comparison of smoothness among four PSO variant algorithms.

A comparative evaluation reveals that while the proposed hybrid architecture achieves a minimized overall trajectory length, it exhibits a marginal degradation in geometric smoothness when contrasted with the QPSO framework. Figure 24 represents the comparison of planning time.

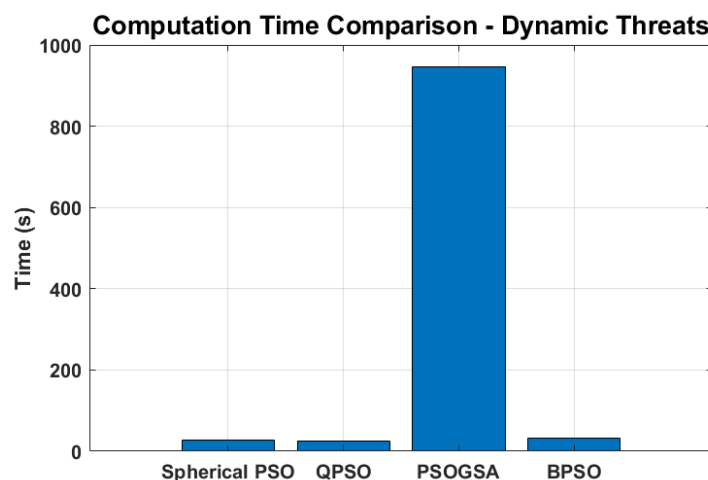


Figure 24. Comparison of planning time among the four PSO variant algorithms.

It can be seen that the planning time of the proposed algorithm is similar to that of the QPSO algorithm. Considering the four metrics comprehensively, the proposed algorithm offers the best overall performance for time-critical rescue missions. In this section, aiming to address the uncertainty of the complex dynamic rescue environment, dynamic patrol threats and sudden radars are introduced, with the focus on evaluating the anti-interference capability of the algorithm. The experiment adopts the robustness metric R defined in Section 4.5 for quantitative scoring.

Figure 25 details the comprehensive performance of the three algorithms under dynamic scenarios, with the robustness score R derived from Equation (52):

$$R = \frac{1}{N} \sum_{i=1}^N (S_{\text{edge}}(i) \times S_{\text{smooth}}(i)) \quad (52)$$

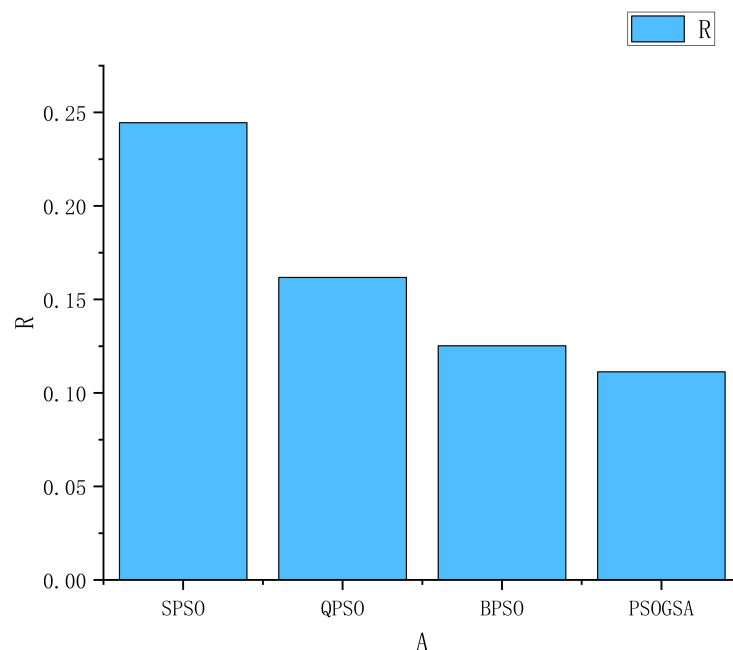


Figure 25. Comparison of robustness metrics R among the four PSO variant algorithms.

Figure 26 presents a radar chart of different algorithms under various metrics. As can be seen in the chart, the fusion algorithm adopted in this paper is superior to the other algorithms in multiple dimensions.

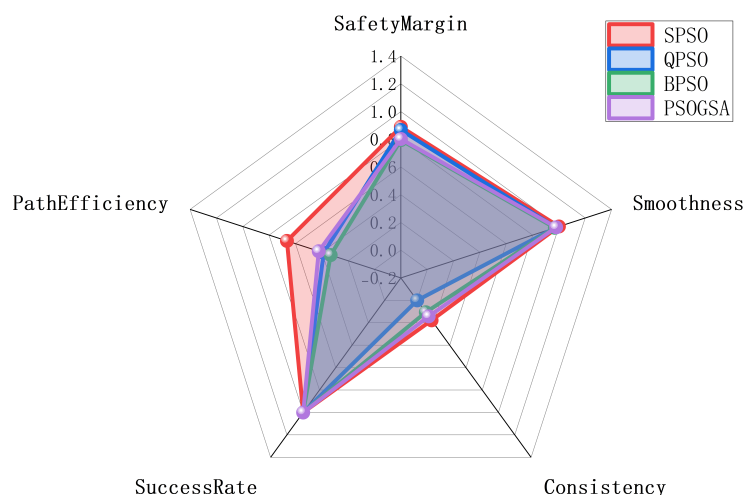


Figure 26. Multi-dimensional Performance Comparison of Different Algorithms.

The experimental data show the following: Conversely, the proposed hybrid architecture intrinsically guarantees a more robust kinematic safety envelope for the synthesized trajectories through the integration of probabilistic diffusion mechanisms within the Gaussian kernel mapping framework. Quantitatively, the mean R metric yielded by the proposed algorithm converges to approximately 0.26, demonstrating a statistically significant superiority over the baseline methodologies, which remained rigidly constrained between 0.1 and 0.16.

The radar chart in Figure 26 further intuitively demonstrates the comprehensive advantages of SPSO in terms of safety, smoothness and consistency, which fully verify the strong robustness of the proposed algorithm in complex dynamic obstacle emergency rescue environments.

Beyond emergency rescue, the dynamic obstacle scenario simulated in Section 6.2 directly mirrors several civil low-altitude operational contexts: the circular-loitering dynamic obstacle models a civil inspection drone conducting facility surveillance, while the linear-transit dynamic obstacle models an express logistics UAV operating on a fixed air corridor. The proposed algorithm's demonstrated capability to predict and avoid both loitering and cruising dynamic obstacles in real time validates its applicability to: (1) urban air mobility (UAM) corridors where multiple heterogeneous aircraft share low-altitude airspace; (2) last-mile logistics delivery where the UAV must navigate around other delivery drones; and (3) infrastructure inspection missions where the primary UAV must yield to patrolling security drones. These connections between the simulation parameters and real-world civil applications address the gap between the paper's stated scope and its experimental validation.

The ablation study aims to quantify the individual contributions of each core component in the improved Spherical Particle Swarm Optimization (SPSO) algorithm. By systematically removing one component at a time and comparing the degraded variant against the full algorithm, we isolate the impact of each improvement on optimization performance. The improved SPSO algorithm incorporates four key components beyond the conventional PSO framework in Table 2.

Table 2. Components of the Spherical Particle Swarm Optimization (SPSO) Algorithm.

Component	Description
Spherical Coordinate Representation	Encodes each waypoint as a spherical vector (r, ψ, ϕ) relative to the previous waypoint, instead of absolute Cartesian coordinates (x, y, z) . This inherently enforces path continuity and compresses the search space.
Directional Guidance	Restricts the azimuth angle ϕ to a narrow cone $(\phi_0 \pm \pi/4)$ centered around the start-to-goal bearing $\phi_0 = \text{atan2}(\Delta y, \Delta x)$, biasing the search toward goal-directed paths.
Velocity Reflection (Mirroring)	When a particle's position exceeds variable bounds, its velocity component is reflected (negated) rather than simply clamped to zero, preserving kinetic energy and improving boundary exploration.
Inertia Weight Damping	The inertia weight w decays multiplicatively each iteration ($w \leftarrow w \times 0.98$), transitioning the swarm from global exploration (high w) to local exploitation (low w).

Five experimental variants were constructed for comparative analysis in Table 3.

Table 3. Ablation Study Variants of the SPSO Algorithm.

Variant	Label	Modification
V1	Full SPSO	Complete algorithm with all four components (baseline).
V2	w/o Spherical	Replaces spherical encoding with direct Cartesian PSO. Each particle optimizes 20 independent (x, y, z) waypoint positions. Initialization uses perturbation around the straight-line path between start and goal. All other components (damping, reflection) are retained.

Table 3. *Cont.*

Variant	Label	Modification
V3	w/o Direction	Removes the directional constraint on ϕ . The azimuth angle is allowed to span the full range $[-\pi, \pi]$ instead of being restricted to $[\phi_0 - \pi/4, \phi_0 + \pi/4]$. All other components remain intact.
V4	w/o Reflection	Disables velocity mirroring at position bounds. When a particle violates a boundary, its position is clamped but the velocity is not negated. All other components remain intact.
V5	w/o Damping	Fixes the inertia weight at $w = 1.0$ throughout all iterations (no decay). All other components remain intact.

Each variant was evaluated using six performance metrics:

1. Mean Cost. Average total cost over 30 independent runs (primary evaluation metric).
2. Cost Standard Deviation—Reflects optimization stability and algorithm robustness.
3. Path Length. Physical distance of the generated trajectory.
4. Computation Time. Actual wall-clock time consumed per run, measured in seconds.
5. Convergence Speed. Iteration number at which 90% of the total cost reduction is attained.
6. Component Contribution. Relative cost degradation defined as $\Delta\% = \frac{C_{\text{variant}} - C_{\text{full}}}{C_{\text{full}}} \times 100\%$.

The parameters of the PSO algorithm were set as follows to reveal the advantage of the spherical coordinate representation, which provides structural priors for path continuity in Table 4.

Table 4. SPSO Algorithm Parameters for Ablation Study.

Parameter	Symbol	Value	Rationale
Number of waypoints	n	20	Increases the decision space to 60 dimensions (20×3 variables), where encoding efficiency becomes critical.
Population size	n_{Pop}	100	A small swarm relative to the 60-dimensional space forces the algorithm to rely on structural priors rather than brute-force coverage.
Maximum iterations	MaxIt	100	A tight iteration budget amplifies the importance of search efficiency and fast convergence.
Monte Carlo repetitions	n_{Runs}	30	Provides statistical robustness for mean and standard deviation estimation.
Random seed	rng($2000 + \text{run}$)	Deterministic per run	Ensures reproducibility and fair comparison across all variants using identical random sequences.

The remaining PSO hyperparameters were held constant across all variants in Table 5:

Table 5. SPSO Algorithm Parameters.

Parameter	Symbol	Value
Initial inertia weight	w_0	1.0
Inertia damping ratio	w_{damp}	0.98
Personal learning coefficient	c_1	1.5
Social learning coefficient	c_2	1.5
Velocity limit ratio	α	0.5
Elevation angle range	ψ	$[-\pi/4, \pi/4]$
Step length upper bound	r_{max}	$2\ S - G\ /n$

The choice of $n = 20$ waypoints with only 100 particles is deliberate. In a 60-dimensional search space, a Cartesian PSO must independently optimize 60 uncoupled variables, with no inherent guarantee of path connectivity. The spherical encoding, by contrast, generates paths incrementally from each waypoint to the next, ensuring continuity by construction. With a limited particle budget, this structural advantage becomes decisive—the spherical representation effectively searches a lower-dimensional manifold of physically plausible paths, while the Cartesian variant wastes particles exploring disconnected or geometrically infeasible configurations. The results are shown in the following Table 6.

Table 6. Parameter settings for the compared algorithms.

Parameter	SPSO	QPSO	BPSO	PSOGSA
MaxIt	200	200	200	200
nPop	500	500	500	500
w	1.0	—	1.0	1.0
w_{damp}	0.98	—	0.98	0.98
c_1	1.5	—	1.5	1.5
c_2	1.5	—	1.5	1.5
Position update	Standard PSO	Quantum (log)	Sigmoid binary	PSO + Gravity

The parameter list comparisons of the four algorithms are provided below to verify the rigor of the algorithm designs. Note that no corresponding parameters are listed for QPSO, as quantum-behaved particle swarm optimization inherently does not involve inertia weights or learning coefficients, which is determined by the fundamental nature of the algorithm itself.

All ant colony optimization (ACO) parameters are kept strictly consistent across all experimental groups, with the specific parameters listed in Table 7 below. However, the only controlled variable in the GKDM ablation experiment is the pheromone initialization strategy. This setup ensures that any observed performance differences can be attributed to variations in the information transfer mechanism.

Table 7. Algorithm parameter settings.

Parameter	Symbol	Value	Description	Rationale
Number of ants	N_a	30	Number of ants per round	commonly 20–50 in ACO literature
Number of iterations	T_{ACO}	50	Total ACO rounds	Consistent with the parameters of PSO
Pheromone exponent	α_p	1.0	Exponent in τ^α	Classic ACO value
Heuristic exponent	β_p	2.0	Exponent in η^β	Classic ACO value
Evaporation rate	ρ	0.08	Pheromone decay per round	commonly 0.05–0.1 in ACO literature
Initial pheromone	τ_0	0.1	Uniform initialization baseline	Classic ACO value

The following are the results of the ablation experiments.

Figure 27 quantitatively compares the mean cost and standard deviation across all variants. The Full SPSO yields the lowest mean cost of 5402.32 ± 382.88 . Removing the spherical coordinate representation (w/o Spherical) results in the most dramatic degradation, with the mean cost nearly doubling to 9900.64 ± 3127.72 —an increase of 83.3%. The w/o Direction variant and w/o Damping variant also show substantial cost increases, confirming the importance of directional guidance and adaptive inertia. The w/o Reflection variant shows only marginal degradation, suggesting that velocity mirroring provides a minor but non-negligible refinement. Notably, the error bar of w/o Spherical is approximately 8× larger than the baseline, indicating that the Cartesian encoding leads to highly unstable optimization under a limited computational budget.

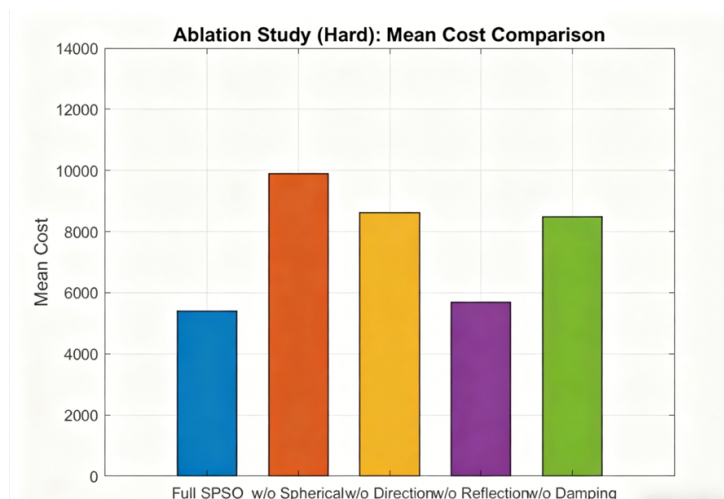


Figure 27. Mean cost comparison of the five ablation variants.

Figure 28 ranks the algorithmic components according to their contributions to the overall optimization performance, where the contribution is quantified by the cost degradation incurred when each component is individually removed. Among them, the spherical coordinate representation makes the most prominent contribution, followed by directional guidance and inertia weight damping, while velocity reflection yields a relatively minor improvement in optimization. This ranking reveals a distinct hierarchy of performance impacts: the representation form of the search space exerts the most significant influence on algorithm performance, especially in high-dimensional scenarios. Directional guidance and inertia damping play complementary roles—the former constrains the spatial search direction, while the latter regulates the balance between global exploration and local exploitation during iterations. Experimental results demonstrate that the spherical-coordinate-based algorithm achieves over 99% performance improvement over the basic Cartesian-coordinate-based algorithm. This analysis provides theoretical guidance for practical applications: when extending path planning algorithms to new problem domains, spherical coordinate encoding and directional guidance should be prioritized as essential core components.

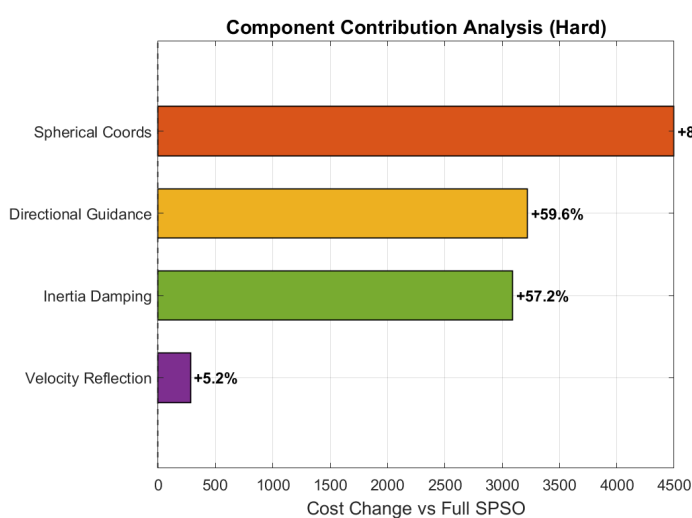


Figure 28. Component contribution analysis.

The ablation experimental results concerning spherical coordinates in this paper reveal that the algorithm cost increases by 83.3% when the spherical coordinate system is discarded under complex scenarios. This performance degradation demonstrates that the

parameterization of continuous waypoints via Cartesian coordinates gives rise to a weakly coupled search space, in which each coordinate (x, y, z) is optimized independently and tends to produce physically inconsistent trajectories. By contrast, the spherical coordinate system couples adjacent waypoints by encoding the vector parameters (r, Ψ, ϕ), thus effectively reducing the search dimensionality.

Figure 29 presents the ablation experimental results of GKDM. The results indicate that the information loss inherent in the common fusion strategies adopted in existing algorithms leads to inferior cost convergence performance. In contrast, the proposed GKDM method achieves more comprehensive information preservation for the PSO-ACO fusion algorithm, enabling thorough information transfer between different algorithms. This performance also demonstrates that, compared with the simple fusion schemes widely used in current research, the GKDM method can effectively improve both the success rate and efficiency of the fusion algorithm. The parameter settings of the GKDM method are listed in Table 8.

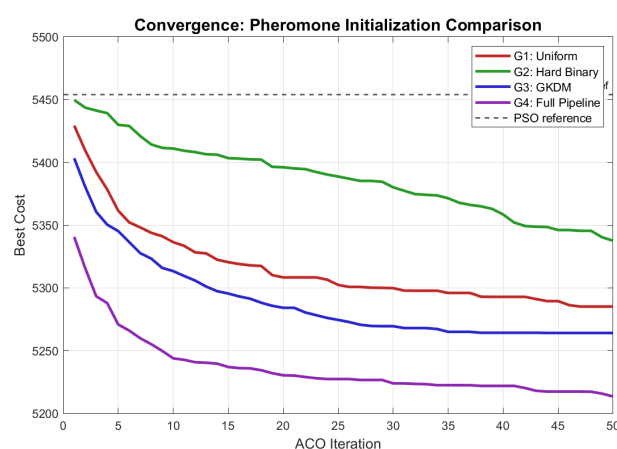


Figure 29. Convergence Plots for Validating the Performance of GKDM.

Table 8. Core parameters of the GKDM-based pheromone initialization.

Symbol	Parameter	Value	Description	Rationale
τ_0	Baseline pheromone	0.1	Ensures global minimum exploration probability	Common value in ACO literature
A	Amplitude coefficient	12	Determines guidance strength of PSO information	Chosen so that $\tau_{\text{peak}} = 1.3$ is of the same order as the Hard Binary peak 3.0
σ	Gaussian kernel bandwidth	40	Effective radius for diffusing continuous PSO information into the discrete space	$3\sigma = 120 \approx \Delta_{xy} = 100$, covering the main region of the search space

6.3. Simulation Experiment in Large-Scale Obstacle-Dense Environment

Authentic large-scale UAV operational environments are rarely limited to a sparse distribution of hazards; rather, macroscopic operational theaters frequently encompass over a hundred distinct threat sources. To rigorously emulate the expansive spatial domains encountered by real-world autonomous aerial vehicles, the simulated operational workspace is scaled to a 3000×3000 coordinate grid. The blue sphere represents spherical obstacles, the red cylinder represents cylindrical obstacles, and the green hemisphere represents hemispherical obstacles. Within this extended Cartesian environment, an ultra-dense array of 100 heterogeneous obstacles—comprising both spherical and cylindrical geometric profiles—is stochastically distributed. The algorithm is run in this scenario for experimental simulation, with the results shown in Figure 30.

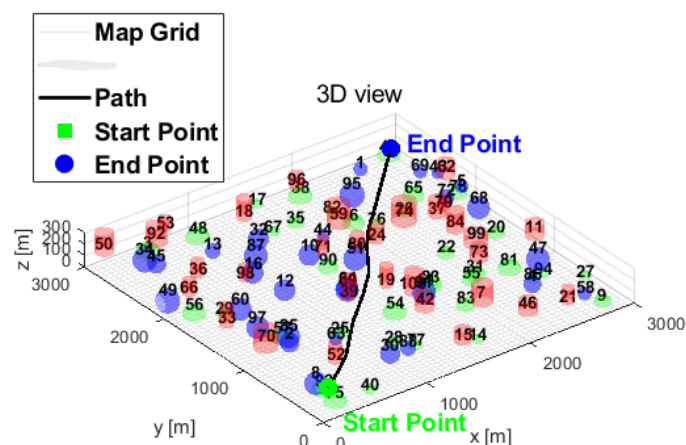


Figure 30. Path Planned by the Improved PSO Algorithm in a Large-scale Operational Area.

Table 9 presents the comprehensive quantitative metrics derived from the UAV trajectory planning simulations utilizing the enhanced SPSO framework. The empirical data unequivocally demonstrate the aircraft's capability to navigate complex spatial hazards while maintaining highly stable kinematic attitudes and smooth trajectory profiles. Ultimately, this enables the system to converge upon the optimal route, thereby successfully executing the designated mission parameters.

Conclusions drawn from Figures 30 and 31 indicate that this algorithm is also applicable in large-scale obstacle-rich environments. By comparing the performance of the traditional PSO algorithm in large-scale obstacle-rich rescue environments, it is concluded that the improved algorithm better meets the needs of practical missions.

Table 9. Path Planning Data in Large-Scale Complex Environments.

Time	X Y Z	Headingdeg	Time	X Y Z	Headingdeg
5	32 38 100	50	190	1285 1399 117	47
10	65 76 100	50	195	1319 1436 118	47
15	97 114 100	50	200	1353 1472 119	47
20	129 152 100	49	205	1387 1509 120	47
25	162 190 100	49	210	1421 1545 121	47
30	195 228 100	49	215	1455 1582 122	47
35	227 266 100	49	220	1489 1618 123	47
40	260 304 100	49	225	1523 1655 124	47
45	293 341 100	48	230	1558 1692 124	47
50	327 379 100	48	235	1592 1728 125	47
55	360 415 100	47	240	1626 1765 126	47
60	394 452 100	47	245	1660 1801 127	47
65	429 489 100	47	250	1694 1838 128	47
70	463 525 100	47	255	1728 1874 129	47
75	497 561 100	46	260	1762 1911 130	47
80	532 597 100	46	265	1797 1947 131	47
85	566 634 100	46	270	1831 1984 132	47
90	601 670 100	46	275	1865 2020 133	47
95	636 706 100	46	280	1899 2057 134	47
100	670 742 100	47	285	1933 2093 135	47
105	704 778 101	47	290	1967 2130 136	47
110	738 815 102	47	295	2001 2166 136	47

Table 9. Cont.

Time	X Y Z	Headingdeg	Time	X Y Z	Headingdeg
115	773 851 103	47	300	2035 2203 137	47
120	807 888 104	47	305	2070 2239 138	47
125	841 924 105	47	310	2104 2276 139	47
130	875 961 106	47	315	2138 2312 140	47
135	909 998 107	47	320	2172 2349 141	47
140	943 1034 108	47	325	2206 2385 142	47
145	977 1071 109	47	330	2240 2422 143	47
150	1011 1107 110	47	335	2274 2459 144	47
155	1046 1144 111	47	340	2308 2495 145	47
160	1080 1180 111	47	345	2343 2532 146	47
165	1114 1217 112	47	350	2377 2568 147	47
170	1148 1253 113	47	355	2411 2605 148	47
175	1182 1290 114	47	360	2445 2641 149	47
180	1216 1326 115	47	365	2479 2678 149	47
185	1250 1363 116	47	368	2500 2700 150	47

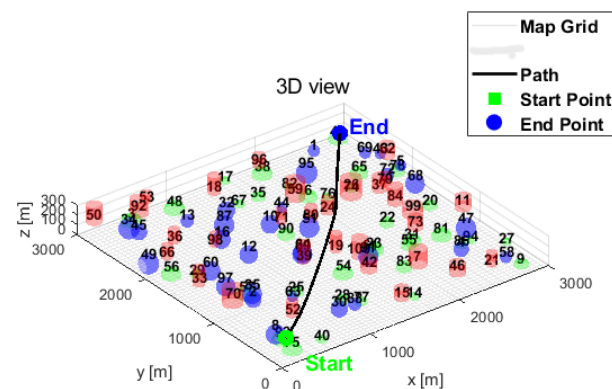


Figure 31. Path Planned by the Traditional PSO Algorithm in a Large-scale Operational Area.

As shown in Figure 32, in terms of cost convergence, the improved PSO algorithm achieves a lower cost value, better satisfying the requirements of practical missions.

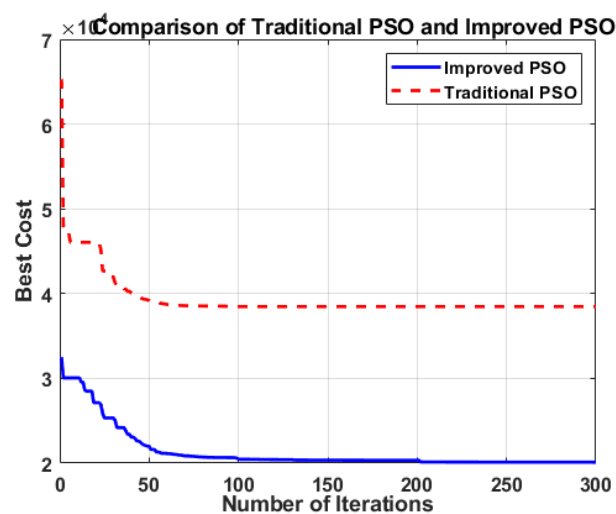


Figure 32. Cost Convergence Comparison of Two PSO Algorithms in a Large-scale Obstacle-Rich Civil Operational Environment.

6.4. Constraint Checking

Following the generation of each trajectory within the emergency rescue simulation, a rigorous post hoc verification protocol is mandated to ascertain compliance with the predefined constraint conditions. While the primary objective cost function was configured to guide the swarm, specific penalty coefficients were deliberately relaxed to facilitate algorithmic convergence and ensure practical aerodynamic execution. Consequently, an independent validation subroutine is integrated into the computational framework. Upon algorithmic termination, the synthesized trajectory is systematically evaluated against four boundaries: (1) spatial collision avoidance, (2) vertical altitude limits, (3) horizontal turning angle thresholds, and (4) longitudinal climb angle restrictions. Ultimately, the cumulative frequency of boundary violations across the entire flight path is quantitatively logged within the output matrix, providing a definitive assessment of the trajectory's overall operational feasibility.

The specific implementation method is as follows[31]:

Regarding the spatial collision constraint: the Euclidean distance between each discrete waypoint along the synthesized trajectory and the proximity of known obstacles is systematically computed. If this calculated distance falls below the predefined minimum safety threshold, a collision constraint violation is formally registered.

Regarding the vertical altitude constraint: the synthesized trajectory is systematically evaluated to identify any flight segments that breach the predefined maximum operational ceiling. If any localized coordinate surpasses this vertical threshold, an altitude constraint violation is formally registered, and the non-compliant segments are graphically isolated for post-simulation analysis.

For the angle constraint: Regarding the horizontal turning angle constraint, the directional vector angle subtended between successive waypoints along the synthesized flight path is rigorously calculated. If this computed angular deviation exceeds the maximum allowable kinematic yaw threshold, an angular constraint violation is formally registered, and the precise spatial coordinates of the non-compliant maneuver are explicitly documented for further analysis[32].

Specifically for emergency rescue applications, the proposed algorithm addresses three domain-specific requirements that generic path planners fail to satisfy simultaneously: (1) real-time replanning capability within the computational budget of on-board rescue UAV processors ; (2) trajectory smoothness that minimizes mechanical stress on payload-carrying rescue drones delivering medical supplies or communication equipment; and (3) robustness to dynamic threat emergence, ensuring mission continuity even when new hazards appear mid-flight.

For the climb angle constraint: Calculate the pitch angle of the line connecting each pair of consecutive points and record any violations.

The output results are shown in Figure 33.

The output results show no constraint violations, indicating that the routes pass the constraint check.

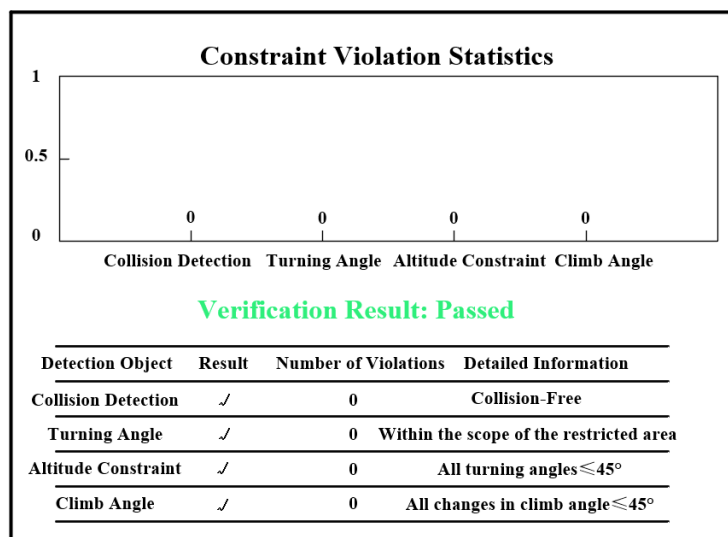


Figure 33. Statistics of Constraint Violations in Route Planning.

7. Conclusions

This study focuses on the route planning problem of rescue UAVs during the ground approach phase in emergency rescue missions and proposes an improved Particle Swarm Optimization (PSO) algorithm. Integrating spherical coordinate system and adaptive weight factor theory, the algorithm can find the global optimal solution in complex emergency rescue environments and generate a visual map of the planned path to support subsequent emergency rescue mission needs.

By transforming the Cartesian search space into a spherical coordinate framework, integrating adaptive inertia weight factors, and formulating a multi-objective cost function tailored to stringent kinematic constraints, the proposed hybrid architecture effectively mitigates the inherent limitations of conventional PSO variants—specifically addressing suboptimal convergence rates and susceptibility to premature local optima stagnation. Extensive simulation validations corroborate that the enhanced algorithm achieves significantly accelerated convergence while reliably isolating the global optimum. Furthermore, within highly complex, obstacle-dense operational theaters—such as seismically active zones, mountainous disaster topographies, and hazardous industrial complexes—the proposed methodology demonstrates marked superiority over established baseline algorithms, yielding unparalleled performance in both flight safety margins and dynamic evasion efficiency. Consequently, this framework is well-suited for complex trajectory planning in disaster reconnaissance, emergency payload delivery, and critical waypoint navigation. Beyond emergency response, the underlying algorithmic architecture possesses immense scalability, allowing for direct extrapolation to civil low-altitude logistics, industrial facility inspection, ecological monitoring, and urban air mobility (UAM) ecosystems. Ultimately, this research offers substantial practical value and robust engineering applicability, fundamentally enhancing the situational awareness, proactive risk mitigation, and autonomous and adaptive navigation capabilities of UAV platforms operating within highly stochastic, low-altitude environments.

Author Contributions: Conceptualization, X.P., X.H. and X.R.; methodology, X.P. and D.Q.; software, X.P. and D.Q.; validation, X.P.; formal analysis, X.P. and X.H.; investigation, X.P.; resources, X.H. and D.Q.; data curation, X.P.; writing—original draft preparation, X.P.; writing—review and editing, X.H.; visualization, X.H. and X.P.; supervision, X.H.; project administration, X.H. All authors have read and agreed to the published version of the manuscript.

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